

ENGINEERING PORTFOLIO PROJECT

Wearable Half-Wave Dipole Antenna

Body Loading, Detuning, Efficiency Degradation, and Retuning

Analytical Design and CST Full-Wave Simulation

Design frequency: 1.120 GHz
Antenna type: Center-fed cylindrical half-wave dipole
Body model: Finite skin-fat-muscle multilayer model
Evidence class: Analytical and simulated
Software: CST Studio Suite Learning Edition and Python

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Portfolio Context

This report documents a complete engineering workflow for a wearable half-wave dipole: analytical initialization, free-space tuning, construction of a simplified lossy multilayer body model, controlled antenna-body spacing studies, radiation-performance comparison, and first-order retuning near the body.

The evidence in this report is **analytical and simulated**. No fabricated prototype, VNA measurement, chamber measurement, or SAR validation is claimed. CST-generated results are reported as simulation outputs for the documented finite geometry, solver settings, material model, and computational boundaries.

The 0.1 mm separation case produced a non-monotonic displacement of the dominant matching minimum. A formal mesh- and boundary-convergence study was not completed because of CST Learning Edition limits. That frequency displacement is therefore classified as a fixed-model observation rather than a convergence-validated trend. The severe target-frequency mismatch, efficiency loss, and gain reduction are retained as the principal engineering observations from the extreme-clearance case.

Native CST project binaries are not included in the public package. Selected figures, model dimensions, simulation notes, analytical scripts, and the complete report source are provided to support technical review and reproducibility within the stated limitations.

Objective

The objective of this project is to design, simulate, and evaluate a wearable half-wave dipole antenna at 1.120 GHz. A center-fed cylindrical PEC dipole is first established and tuned in free space. The same antenna is then evaluated above a simplified finite multilayer body model composed of skin, fat, and muscle.

The study quantifies how antenna-body separations of 10 mm, 5 mm, and 0.1 mm affect the reflection coefficient S_{11} , the dominant matching-minimum frequency, radiation efficiency, total efficiency, gain, and two-dimensional and three-dimensional radiation patterns. A first-order redesign is then performed for the 5 mm case by shortening the dipole and evaluating a feed-gap variation.

The scope is electromagnetic antenna performance. The finite tissue model is used as a lossy loading environment; biological safety and SAR compliance are outside the scope of this report.

Abstract

This project investigates a wearable center-fed half-wave dipole operating at 1.120 GHz near a simplified finite skin-fat-muscle body model. The theoretical free-space half-wave length was 133.93 mm. Full-wave tuning reduced the total dipole length to 122.0 mm, yielding a minimum S_{11} of -15.52702 dB at 1.1168 GHz and $S_{11}(1.120 \text{ GHz}) = -15.48433$ dB.

Body loading degraded both impedance matching and radiating performance. At separations of 10 mm and 5 mm, the dominant matching minima shifted to 1.0312 GHz and 0.9792 GHz, while the target-frequency reflection coefficients degraded to -7.839072 dB and -5.903831 dB, respectively. At 0.1 mm, the documented model produced a dominant minimum at 1.3976 GHz and $S_{11}(1.120 \text{ GHz}) = -6.951098$ dB. Because the non-monotonic frequency displacement was not verified through mesh or boundary convergence, it is reported as a fixed-model observation rather than a validated physical trend.

At the design frequency, gain decreased from 0.7515 dBi at 10 mm to 0.3534 dBi at 5 mm, and to -5.261 dBi at 0.1 mm. Shortening the dipole to 106.6 mm for the 5 mm case improved $S_{11}(1.120 \text{ GHz})$ from -5.903831 dB to -8.000936 dB, but did not recover the common -10 dB criterion. The study shows that wearable antenna design requires simultaneous control of spacing, detuning, mismatch, tissue absorption, efficiency, and gain.

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Chapter 1

Wearable Half-Wave Dipole Study

1.1 Introduction

Wearable antennas are widely used in wireless body-area networks, biomedical monitoring systems, portable communication devices, and personal wireless systems. Unlike an isolated antenna in free space, a wearable antenna operates in the immediate vicinity of the human body. Since biological tissues are lossy and have relatively high dielectric constants, the nearby body can significantly affect the antenna input impedance, resonant frequency, radiation efficiency, gain, and radiation pattern.

The purpose of this project is to design and analyze a wearable half-wave dipole antenna using CST Studio Suite. The antenna is first designed and tuned in free space at the design operating frequency of

$$f_0 = 1.120 \text{ GHz.}$$

This free-space model provides a baseline response for the antenna before introducing the human body model. After the free-space design is verified, the dipole antenna is placed above an approximate multilayer human body model consisting of skin, fat, and muscle layers. The antenna performance is then evaluated for three separation distances between the antenna and the body: 10 mm, 5 mm, and 0.1 mm.

The main objective is to investigate how the presence and proximity of the lossy body model affect the wearable antenna. The quantities of interest include the return loss S_{11} , resonant frequency shift, radiation efficiency, total efficiency, gain, and radiation patterns. The project also includes a retuning step to examine whether the degraded impedance matching can be improved after the antenna is placed near the body.

It is important to clarify the scope of this project. This work is not intended to be a biological safety study, a specific absorption rate (SAR) analysis, or a complete RF-exposure assessment. Instead, the objective is to study the effect of the human body on antenna performance. The body is treated as a lossy electromagnetic loading environment that perturbs the antenna input impedance and modifies the radiated field behavior.

The simulation results demonstrate that the nearby body significantly detunes the antenna and reduces its radiating performance. As the antenna is moved closer to the body, the interaction becomes stronger because the body is located in the antenna near-field region. This near-field coupling changes the stored electromagnetic energy around the antenna, increases power absorption in the lossy tissue layers, and reduces the effective radiated power. The far-field radiation patterns reported in this project are obtained from CST far-field monitors, which compute the far-zone radiation behavior of the combined antenna-body system after solving the near-field electromagnetic interaction.

1.2 Design Specifications

The design operating frequency was selected as

$$f_0 = 1.120 \text{ GHz.}$$

This value defines the wavelength scale, the initial half-wave estimate, the simulation sweep, and the far-field monitor frequency used throughout the study.

The corresponding free-space wavelength is

$$\lambda_0 = \frac{c}{f_0},$$

where $c \approx 3.0 \times 10^8 \text{ m/s}$ is the speed of light in free space. Therefore,

$$\lambda_0 = \frac{3.0 \times 10^8}{1.120 \times 10^9} = 0.26786 \text{ m} = 267.86 \text{ mm.}$$

The theoretical free-space half-wavelength dipole length is obtained from the standard half-wave dipole approximation [1],

$$L_{\lambda/2} = \frac{\lambda_0}{2} = 133.93 \text{ mm.}$$

This value was used as the initial physical length of the dipole antenna before CST tuning.

The corresponding Python verification is provided in Appendix A, Listing A.1.

The cylindrical dipole wire diameter was specified as

$$d = 1.81 \text{ mm.}$$

The corresponding wire radius used in the CST cylindrical dipole model is

$$a = \frac{d}{2} = 0.905 \text{ mm.}$$

The antenna was modeled as a center-fed cylindrical PEC dipole aligned along the x -axis. A 2 mm feed gap was used at the center of the dipole, and a discrete port with 50Ω reference impedance was placed across the gap. This geometry provides a practical full-wave model of the specified wire-dipole antenna configuration.

Table 1.1: Initial dipole design parameters.

Parameter	Value
Target operating frequency, f_0	1.120 GHz
Free-space wavelength, λ_0	267.86 mm
Initial half-wave dipole length, $L_{\lambda/2}$	133.93 mm
Wire diameter, d	1.81 mm
Wire radius, a	0.905 mm
Feed gap	2.00 mm
Port reference impedance	50Ω
Dipole orientation	x -axis

The approximate human body model was constructed using three stacked lossy dielectric layers representing skin, fat, and muscle. The tissue material properties were selected to represent the lossy dielectric behavior of biological tissues at microwave frequencies [2]. The layer thicknesses were kept unchanged in all body-separation simulations.

Table 1.2: Multilayer human body material model used in CST.

Layer	Relative Permittivity	Conductivity (S/m)	Thickness (mm)
Skin	41.32	0.855	1.5
Fat	5.46	0.050	20.0
Muscle	54.97	0.934	30.0

Due to the mesh-cell limitation of the CST Studio Suite Learning Edition, the body was modeled as a finite rectangular tissue patch rather than a full anatomical body model. The final body patch dimensions used in the simulations were

$$150 \text{ mm} \times 60 \text{ mm},$$

with the three tissue layers stacked in the z -direction. The tissue thicknesses, material properties, antenna geometry, and antenna-body separation distances were held fixed across the comparative simulations.

Table 1.3: Geometrical dimensions of the finite multilayer body patch.

Parameter	Value
Body patch length in x -direction	150 mm
Body patch width in y -direction	60 mm
Skin thickness	1.5 mm
Fat thickness	20.0 mm
Muscle thickness	30.0 mm
Total tissue thickness	51.5 mm

The antenna was simulated in four main configurations: free space, 10 mm above the body, 5 mm above the body, and 0.1 mm above the body. The separation distance was defined as the air gap between the lower surface of the dipole wire and the top surface of the skin layer.

Table 1.4: Simulation cases considered in this project.

Simulation Case	Antenna-Body Separation
Free-space dipole	No body model
Body case 1	10 mm
Body case 2	5 mm
Body case 3	0.1 mm
Retuned body case	5 mm

For all simulations, the frequency range was set from 0.7 GHz to 1.5 GHz. The full-wave simulations were performed in CST Studio Suite using the time-domain solver, with a far-field monitor defined at 1.120 GHz [3]. The main extracted results were the return loss S_{11} , resonant or dominant matching-minimum frequency, radiation efficiency, total efficiency, gain, and two-dimensional and three-dimensional radiation patterns.

1.3 Theoretical Background

The CST results in this project are interpreted using several fundamental antenna concepts, including half-wave dipole behavior, impedance matching, radiation efficiency, gain, lossy dielectric loading, and near-field coupling [1], [4]. These concepts provide the analytical basis for understanding why the antenna response changes when the dipole is moved from free space to the vicinity of the multilayer body model.

1.3.1 Free-Space Half-Wave Dipole

A center-fed half-wave dipole is a resonant wire antenna whose total length is approximately one half of the operating wavelength. For a free-space operating frequency of

$$f_0 = 1.120 \text{ GHz},$$

the free-space wavelength is

$$\lambda_0 = \frac{c}{f_0} = 267.86 \text{ mm}.$$

The corresponding theoretical half-wavelength is

$$L_{\lambda/2} = \frac{\lambda_0}{2} = 133.93 \text{ mm}.$$

This value provides the initial design estimate. In practice, the resonant length of a finite-radius dipole with a finite feed gap is usually different from exactly $\lambda_0/2$. Therefore, CST tuning is required to place the minimum return loss close to the desired operating frequency.

The ideal radiation behavior of a half-wave dipole is broadside to the wire axis. For a dipole oriented along the x -axis, the strongest radiation is expected in directions perpendicular to the dipole axis. In free space, a thin half-wave dipole has a directivity of approximately

$$D \approx 1.64,$$

or

$$D_{\text{dBi}} \approx 2.15 \text{ dBi}.$$

This value provides a useful reference for evaluating the CST radiation-pattern results.

1.3.2 Return Loss and Impedance Matching

The return loss behavior of the antenna is described by the reflection coefficient S_{11} . For a one-port antenna connected to a transmission line with reference impedance Z_0 , the reflection coefficient is related to the antenna input impedance Z_{in} by

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}.$$

The magnitude of S_{11} in decibels is

$$S_{11,\text{dB}} = 20 \log_{10} |\Gamma|.$$

A more negative value of S_{11} indicates better impedance matching and less reflected power. A commonly used practical design criterion is

$$S_{11} < -10 \text{ dB},$$

which corresponds to less than about 10% of the incident power being reflected. In this project, the target is to obtain good matching near

$$1.120 \text{ GHz}.$$

When the antenna is placed near the body model, the input impedance changes because the body modifies the near electric and magnetic fields around the dipole. Therefore, the return loss curve may shift, broaden, weaken, or develop a different dominant matching minimum.

1.3.3 Radiation Efficiency, Total Efficiency, and Gain

Directivity describes only how strongly the antenna concentrates radiation in a particular direction. Gain includes the effect of radiation efficiency. If D is the directivity and e_r is the radiation efficiency, then

$$G = e_r D.$$

In decibel form,

$$G_{\text{dBi}} = D_{\text{dBi}} + e_{r,\text{dB}}.$$

Radiation efficiency accounts for power lost in the antenna structure and nearby lossy materials. For a wearable antenna, the nearby human body can absorb part of the accepted power because biological tissues have nonzero conductivity. This reduces the radiated power and therefore lowers the antenna gain.

CST also reports total efficiency. Total efficiency includes both radiation loss and mismatch loss. Therefore, total efficiency is generally lower than radiation efficiency when the antenna is not well matched at the operating frequency.

1.3.4 Lossy Dielectric Loading by the Human Body

The human body model in this project is represented by three lossy dielectric layers: skin, fat, and muscle. Each tissue is characterized by a relative permittivity ε_r and conductivity σ . For a time-harmonic field with angular frequency

$$\omega = 2\pi f,$$

a lossy dielectric can be represented using the complex permittivity

$$\varepsilon_c = \varepsilon_0 \varepsilon_r - j \frac{\sigma}{\omega},$$

or equivalently,

$$\varepsilon_{r,c} = \varepsilon_r - j \frac{\sigma}{\omega \varepsilon_0}.$$

The real part of the permittivity stores electric energy, while the conductivity term represents dissipative loss. Since skin and muscle have high relative permittivity and significant conductivity, they strongly affect the electromagnetic fields when placed near the dipole.

When the body is close to the antenna, the effective electromagnetic environment around the dipole is no longer free space. The high-permittivity tissue increases the

effective electrical loading of the antenna. This can change the effective electrical length of the dipole and shift the input-matching behavior. In addition, tissue conductivity absorbs electromagnetic power, reducing radiation efficiency and gain.

1.3.5 Near-Field Coupling and Far-Field Radiation Pattern

A key point in this project is the distinction between antenna-body separation and far-field radiation-pattern observation. The body model is located very close to the antenna. At

$$f_0 = 1.120 \text{ GHz},$$

the free-space wavelength is

$$\lambda_0 = 267.86 \text{ mm}.$$

Therefore, the three antenna-body separation distances correspond to

$$10 \text{ mm} \approx 0.037\lambda_0,$$

$$5 \text{ mm} \approx 0.019\lambda_0,$$

and

$$0.1 \text{ mm} \approx 0.00037\lambda_0.$$

Thus, the body is located in the near-field region of the antenna for all three cases. This is especially true for the 0.1 mm case, where the body is almost touching the dipole and strongly interacts with the reactive near fields.

For a finite antenna with largest dimension D , one common far-field distance criterion is

$$R_{\text{ff}} \approx \frac{2D^2}{\lambda_0}.$$

Using the tuned free-space dipole length of approximately

$$D \approx 122 \text{ mm},$$

this gives

$$R_{\text{ff}} \approx \frac{2(122 \text{ mm})^2}{267.86 \text{ mm}} \approx 111 \text{ mm}.$$

Other conservative criteria, such as $5D$ or $1.6\lambda_0$, give even larger far-field distances. Therefore, the body distances of 10 mm, 5 mm, and 0.1 mm are not far-field distances. They are near-field coupling distances.

This does not contradict the use of a CST far-field monitor. The CST solver first computes the electromagnetic fields around the complete antenna-body structure. The far-field monitor then calculates the far-zone radiation pattern of the combined antenna-body system using a near-to-far-field transformation. Therefore, the body can be in the near field of the antenna while the reported radiation pattern still represents the far-field radiation behavior of the complete structure.

1.3.6 Expected Effect of Body Proximity

As the dipole is moved closer to the body model, several effects are expected:

- the antenna input impedance changes because of near-field coupling,
- the return-loss minimum shifts away from the free-space tuned value,
- the S_{11} value at the target frequency may degrade,
- part of the accepted power is absorbed in the lossy tissues,
- radiation efficiency decreases,
- gain decreases,
- the radiation pattern can become distorted by the nearby body.

For moderate separation distances, the high-permittivity body loading often increases the effective electrical length of the antenna and can shift the matching minimum downward in frequency. However, for extremely small separation distances, such as 0.1 mm, the interaction becomes a strong coupled antenna-body problem. In that case, the S_{11} response does not necessarily follow a simple monotonic resonance shift. Instead, the dominant matching minimum may move to a different frequency branch because the input impedance is strongly perturbed by reactive near-field coupling and tissue loss.

1.3.7 Retuning Strategy

After the body is introduced, the original free-space-tuned dipole no longer remains optimally matched at the target frequency. A first-order retuning estimate can be obtained by assuming that the resonant frequency is approximately inversely proportional to the dipole length:

$$f_r \propto \frac{1}{L}.$$

Therefore, if the original dipole length is L_{old} , the observed matching-minimum frequency is f_r , and the desired target frequency is f_0 , then the estimated retuned length is

$$L_{\text{new}} = L_{\text{old}} \frac{f_r}{f_0}.$$

This relation provides only an initial estimate. Near the lossy body, the antenna input impedance is affected by tissue loading, finite geometry, conductor radius, feed gap, and boundary conditions. Therefore, length-only retuning may improve the response but may not fully recover the desired $S_{11} < -10$ dB matching condition. A complete wearable antenna redesign may require additional tuning of feed gap, conductor radius, antenna height, or an impedance-matching network.

1.4 Free-Space Dipole Design and Tuning

Before studying the effect of the human body, the dipole antenna must first be established as a valid free-space baseline. The initial geometry was based on the theoretical half-wavelength estimate, and the CST model was then tuned so that the return-loss minimum occurred close to the design operating frequency of 1.120 GHz.

1.4.1 Initial Free-Space Geometry

The wearable antenna was first designed and simulated in free space before introducing the human body model. The initial dipole length was selected from the theoretical half-wavelength value at the target frequency,

$$L_{\lambda/2} = 133.93 \text{ mm}.$$

The dipole was modeled as two PEC cylindrical arms separated by a 2 mm center feed gap. The wire radius was

$$a = 0.905 \text{ mm.}$$

Because the dipole was oriented along the x -axis, the initial arm length was calculated as

$$L_{\text{arm}} = \frac{L_{\lambda/2} - g}{2},$$

where $g = 2 \text{ mm}$ is the feed gap. Therefore,

$$L_{\text{arm}} = \frac{133.93 - 2}{2} = 65.965 \text{ mm.}$$

The right and left dipole arms were defined symmetrically about the origin. The discrete port was placed across the center gap.

Table 1.5: Initial free-space dipole geometry in CST.

Parameter	Value
Total dipole length	133.93 mm
Feed gap	2.00 mm
Wire radius	0.905 mm
Each arm length	65.965 mm
Right arm x -range	1.000 mm to 66.965 mm
Left arm x -range	−66.965 mm to −1.000 mm
Port reference impedance	50 Ω

The initial CST simulation was performed over the frequency range from 0.7 GHz to 1.5 GHz. The purpose of this first simulation was to check whether the theoretical half-wavelength estimate produced a good impedance match at the design frequency of 1.120 GHz.

1.4.2 Initial Free-Space Result

The initial free-space simulation showed that the antenna was resonant below the target frequency. The minimum return loss occurred at approximately

$$f_{\text{min}} = 1.020 \text{ GHz,}$$

with

$$S_{11,\min} = -15.630935 \text{ dB.}$$

However, at the target frequency,

$$f_0 = 1.120 \text{ GHz,}$$

the return loss was only

$$S_{11}(1.120 \text{ GHz}) = -6.45269 \text{ dB.}$$

This result indicates that the theoretical half-wavelength length was electrically too long for the practical CST model. The difference is expected because the actual antenna includes finite wire radius, a finite feed gap, and a discrete feed model. Therefore, the physical resonant length is not exactly equal to $\lambda_0/2$.

Table 1.6: Initial free-space CST result before tuning.

Quantity	Value
Initial dipole length	133.93 mm
Target frequency	1.120 GHz
Minimum- S_{11} frequency	1.020 GHz
Minimum S_{11}	-15.630935 dB
S_{11} at 1.120 GHz	-6.45269 dB

The initial free-space return-loss response before length tuning is shown in Figure 1.1.

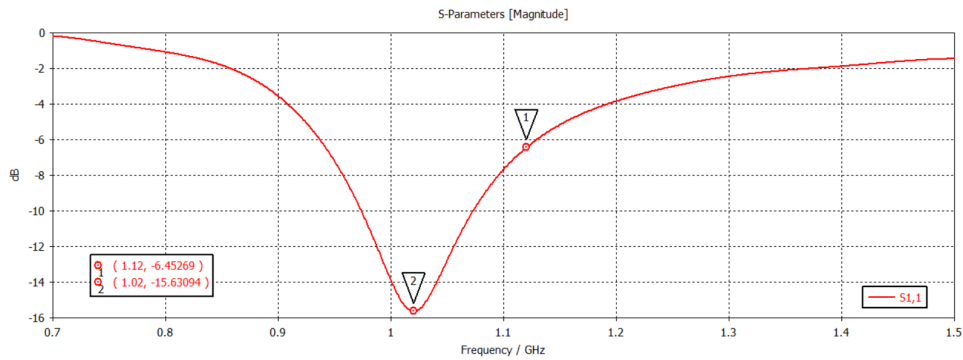


Figure 1.1: Initial free-space return-loss response for the theoretical half-wave dipole length $L = 133.93$ mm. The minimum occurs at 1.020 GHz, while S_{11} at 1.120 GHz is -6.45269 dB.

The initial simulation therefore provided the direction for tuning: since the antenna resonated below the target frequency, the dipole length needed to be shortened.

1.4.3 Length Tuning

A first-order tuning estimate was obtained using the approximate inverse relation between resonant frequency and dipole length,

$$f_r \propto \frac{1}{L}.$$

The tuned length was estimated from

$$L_{\text{new}} = L_{\text{old}} \frac{f_r}{f_0}.$$

Using

$$L_{\text{old}} = 133.93 \text{ mm},$$

$$f_r = 1.020 \text{ GHz},$$

and

$$f_0 = 1.120 \text{ GHz},$$

gives

$$L_{\text{new}} = 133.93 \frac{1.020}{1.120} = 121.99 \text{ mm} \approx 122.0 \text{ mm}.$$

This first-order free-space length-retuning calculation is verified using the Python utility script in Appendix A, Listing A.1.

Therefore, the tuned free-space dipole length was selected as

$$L_{\text{tuned}} = 122.0 \text{ mm}.$$

With the same 2 mm feed gap, the tuned arm length became

$$L_{\text{arm,tuned}} = \frac{122.0 - 2.0}{2} = 60.0 \text{ mm.}$$

Thus, the tuned CST geometry was defined using the following coordinates.

Table 1.7: Tuned free-space dipole geometry in CST.

Parameter	Value
Tuned total dipole length	122.0 mm
Feed gap	2.0 mm
Wire radius	0.905 mm
Each arm length	60.0 mm
Right arm x -range	1.0 mm to 61.0 mm
Left arm x -range	-61.0 mm to -1.0 mm
Discrete port location	$x = -1.0 \text{ mm}$ to $x = 1.0 \text{ mm}$
Port reference impedance	50 Ω

The tuned free-space geometry used as the baseline model is shown in Figure 1.2.

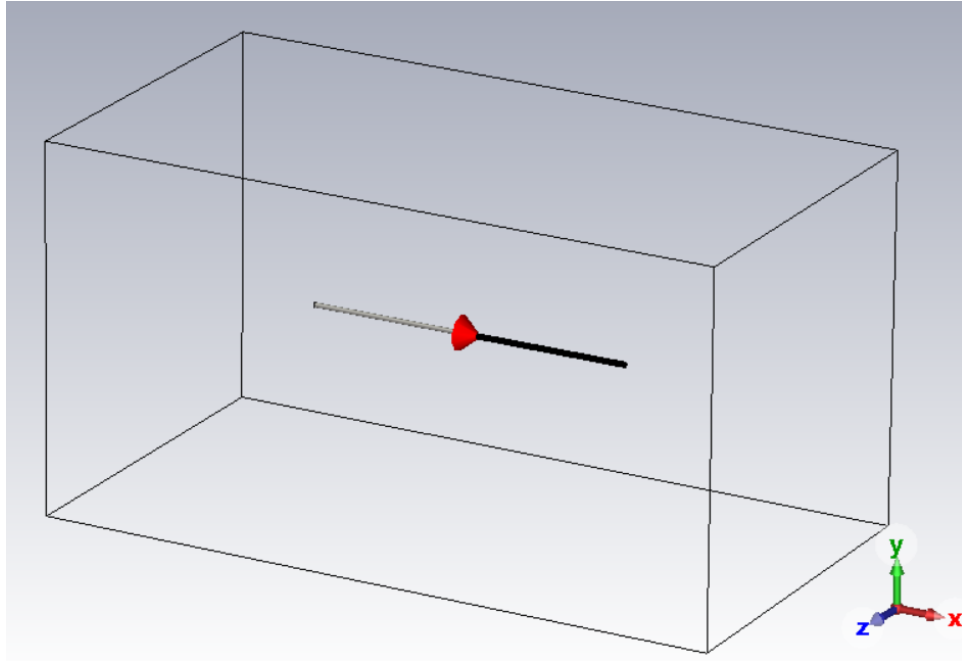


Figure 1.2: Tuned free-space dipole geometry with total length $L = 122 \text{ mm}$, wire radius 0.905 mm , and 2 mm center feed gap.

1.4.4 Tuned Free-Space Result

After shortening the dipole to 122.0 mm, the return-loss minimum moved close to the design target frequency. The tuned free-space result was

$$f_{\min} = 1.1168 \text{ GHz},$$

with

$$S_{11,\min} = -15.52702 \text{ dB}.$$

At the target frequency,

$$S_{11}(1.120 \text{ GHz}) = -15.48433 \text{ dB}.$$

This confirms that the tuned free-space dipole is well matched near 1.120 GHz, since the return loss is below the commonly used -10 dB matching criterion.

Table 1.8: Tuned free-space CST result.

Quantity	Value
Tuned dipole length	122.0 mm
Target frequency	1.120 GHz
Minimum- S_{11} frequency	1.1168 GHz
Minimum S_{11}	-15.52702 dB
S_{11} at 1.120 GHz	-15.48433 dB

The corresponding tuned free-space return-loss response is shown in Figure 1.3.

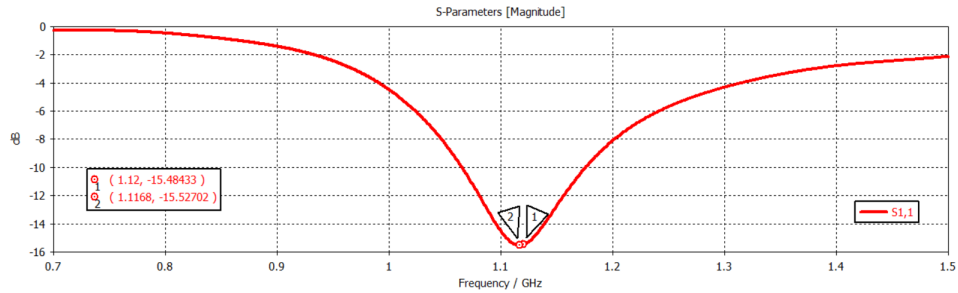


Figure 1.3: Return loss of the tuned free-space dipole. The minimum occurs at 1.1168 GHz, and S_{11} at the target frequency 1.120 GHz is -15.48433 dB.

The tuned free-space antenna was used as the baseline design for all subsequent body-loading simulations. This baseline is important because any later change in S_{11} , gain,

radiation efficiency, or radiation pattern can be attributed to the presence and proximity of the lossy multilayer body model.

1.4.5 Free-Space Baseline Interpretation

The free-space simulation demonstrates an important practical antenna-design point. Although the theoretical half-wavelength value gives a useful starting point, the final resonant length must be adjusted in the full-wave model. The final tuned length,

$$L_{\text{tuned}} = 122.0 \text{ mm},$$

is shorter than the theoretical value,

$$L_{\lambda/2} = 133.93 \text{ mm}.$$

The percentage reduction is

$$\frac{133.93 - 122.0}{133.93} \times 100 = 8.91\%.$$

This shortening is consistent with the fact that finite-radius dipoles and practical feed models do not resonate exactly at the ideal $\lambda_0/2$ length. Once tuned, the antenna achieved good matching at the design frequency and could be used as a valid reference for the wearable antenna study.

1.5 Multilayer Human Body Model

To evaluate wearable antenna behavior, the free-space environment was replaced by a simplified lossy body-loading environment. The model uses three planar tissue layers to represent skin, fat, and muscle, allowing the main electromagnetic effects of high permittivity and tissue conductivity to be included while keeping the simulation feasible in the CST Learning Edition.

1.5.1 Modeling Approach

The human body was approximated as a finite rectangular multilayer dielectric block. This simplified geometry was selected because it captures the main electromagnetic

effect of the body near a wearable antenna while remaining computationally feasible in the CST Studio Suite Learning Edition.

The body model was placed below the dipole antenna. The dipole remained centered at

$$z = 0,$$

and the tissue layers were stacked in the negative z -direction. The antenna-body separation distance was defined as the air gap between the lower surface of the cylindrical dipole wire and the top surface of the skin layer.

Since the dipole wire radius was

$$a = 0.905 \text{ mm},$$

the lower surface of the dipole wire was located at

$$z = -0.905 \text{ mm}.$$

Therefore, for a specified air gap d , the top surface of the skin layer was placed at

$$z_{\text{skin,top}} = -(d + a).$$

This definition was used consistently for the 10 mm, 5 mm, and 0.1 mm body-separation cases.

1.5.2 Tissue Material Properties

The body model consisted of skin, fat, and muscle layers. Each layer was modeled as a lossy dielectric material with the relative permittivity and conductivity specified in the design specification. The material parameters used in CST are summarized in Table 1.9.

Table 1.9: Tissue material properties used in the multilayer body model.

Tissue Layer	ϵ_r	σ (S/m)	Thickness (mm)
Skin	41.32	0.855	1.5
Fat	5.46	0.050	20.0
Muscle	54.97	0.934	30.0

The use of frequency-dependent permittivity and conductivity is essential because biological tissues behave as lossy dielectric media at microwave frequencies [2].

The high relative permittivity of skin and muscle increases the electromagnetic loading near the dipole, while the nonzero conductivity of the tissues introduces dissipative loss. Therefore, when the antenna is placed near the body, the tissue layers can detune the antenna and reduce its radiation efficiency and gain.

1.5.3 Finite Body Patch Dimensions

Due to the 100,000-cell mesh limitation of the CST Studio Suite Learning Edition, the body was modeled as a finite rectangular patch instead of a full-size anatomical body. The final body patch dimensions used in the simulations were

$$150 \text{ mm} \times 60 \text{ mm},$$

with the length along the x -direction and the width along the y -direction. The three tissue layers were stacked along the z -direction.

The coordinate limits of the body patch were

$$-75 \text{ mm} \leq x \leq 75 \text{ mm},$$

and

$$-30 \text{ mm} \leq y \leq 30 \text{ mm}.$$

The tissue-layer thicknesses were kept unchanged from the model specification.

Table 1.10: Finite body patch geometry used in CST.

Parameter	Value
Body patch length in x -direction	150 mm
Body patch width in y -direction	60 mm
Skin thickness	1.5 mm
Fat thickness	20.0 mm
Muscle thickness	30.0 mm
Total tissue thickness	51.5 mm

This finite body patch is large enough to represent the local lossy loading environment directly beneath the dipole while keeping the total mesh size within the CST Learning Edition limit.

1.5.4 Layer Coordinates for the 10 mm Case

For the 10 mm separation case, the top of the skin layer was located at

$$z_{\text{skin,top}} = -(10 + 0.905) = -10.905 \text{ mm.}$$

The skin, fat, and muscle layers were then stacked downward. The resulting z -coordinates are summarized in Table 1.11.

Table 1.11: Body-layer z -coordinates for the 10 mm antenna-body separation case.

Layer	$z_{\min}$ (mm)	$z_{\max}$ (mm)
Skin	-12.405	-10.905
Fat	-32.405	-12.405
Muscle	-62.405	-32.405

1.5.5 Layer Coordinates for the 5 mm Case

For the 5 mm separation case, the top of the skin layer was located at

$$z_{\text{skin,top}} = -(5 + 0.905) = -5.905 \text{ mm.}$$

The corresponding tissue-layer coordinates are listed in Table 1.12.

Table 1.12: Body-layer z -coordinates for the 5 mm antenna-body separation case.

Layer	$z_{\min}$ (mm)	$z_{\max}$ (mm)
Skin	-7.405	-5.905
Fat	-27.405	-7.405
Muscle	-57.405	-27.405

1.5.6 Layer Coordinates for the 0.1 mm Case

For the 0.1 mm separation case, the top of the skin layer was placed very close to the lower surface of the dipole wire. The skin top coordinate was

$$z_{\text{skin,top}} = -(0.1 + 0.905) = -1.005 \text{ mm.}$$

The corresponding tissue-layer coordinates are shown in Table 1.13.

Table 1.13: Body-layer z -coordinates for the 0.1 mm antenna-body separation case.

Layer	$z_{\min}$ (mm)	$z_{\max}$ (mm)
Skin	-2.505	-1.005
Fat	-22.505	-2.505
Muscle	-52.505	-22.505

This case represents the strongest near-field coupling condition because the lossy tissue model is almost in contact with the dipole wire.

1.5.7 CST Learning Edition Mesh-Limit Consideration

The simulations were performed using the CST Studio Suite Learning Edition. This version limits the maximum number of mesh cells to 100,000. The initial full computational domain exceeded this mesh-cell limit after the multilayer body model was added.

To make the simulations feasible while preserving the essential model parameters, the open boundary spacing was reduced. The boundary type was kept as open add space, but the automatic minimum distance to the structure was changed to an absolute distance of

$$10 \text{ mm.}$$

The estimated reflection level was set to

$$10^{-3}.$$

The tissue material properties, tissue thicknesses, antenna geometry, target frequency, and antenna-body separation distances were not changed.

CST reported a reduced-PML-layer warning because the computational domain was intentionally reduced to satisfy the Learning Edition mesh limit. However, the time-domain solver completed successfully and reached the steady-state energy criterion. Therefore, the results were accepted for comparative engineering analysis among the free-space, 10 mm, 5 mm, and 0.1 mm separation cases.

1.5.8 Purpose of the Body Model

The body model was used to study the effect of a nearby lossy dielectric structure on wearable antenna performance. The quantities evaluated in the following sections include S_{11} , resonant-frequency shift, radiation efficiency, total efficiency, gain, and radiation patterns.

The simplified multilayer model is appropriate for this project because the main engineering objective is to observe how the antenna response changes when the free-space environment is replaced by a high-permittivity and lossy body-loading environment. A full anatomical body simulation or SAR assessment would require a more detailed model and is outside the scope of this project.

1.6 Dipole Antenna Above the Multilayer Body Model

After the free-space antenna was tuned, the same dipole was placed above the multilayer body model at three different separation distances. Comparing these cases shows how increasing near-field coupling to the lossy body changes the antenna impedance, matching frequency, efficiency, gain, and radiation behavior.

1.6.1 Body Separation of 10 mm

The 10 mm separation case represents the weakest body-loading condition in this study. Although this distance is the largest spacing considered, it is still electrically small compared with the free-space wavelength at 1.120 GHz. Therefore, measurable changes in impedance matching, efficiency, gain, and radiation pattern are still expected due to near-field coupling between the dipole and the lossy multilayer body model.

Geometry and Simulation Setup

The tuned free-space dipole length was kept fixed at

$$L = 122.0 \text{ mm.}$$

The wire radius was

$$a = 0.905 \text{ mm},$$

and the feed gap remained

$$g = 2.0 \text{ mm}.$$

For the 10 mm body-separation case, the air gap was defined as the distance between the lower surface of the cylindrical dipole wire and the top surface of the skin layer. Since the lower surface of the wire is located at

$$z = -0.905 \text{ mm},$$

the top of the skin layer was placed at

$$z_{\text{skin,top}} = -(10 + 0.905) = -10.905 \text{ mm}.$$

The skin, fat, and muscle layers were then stacked downward in the negative z -direction. The tissue coordinates for this case were already summarized in Table 1.11. The same material properties, layer thicknesses, boundary settings, and frequency range were used as described in the previous section.

The CST time-domain solver was used over the frequency range

$$0.7 \text{ GHz} \leq f \leq 1.5 \text{ GHz}.$$

A far-field monitor was defined at the target frequency,

$$f_0 = 1.120 \text{ GHz}.$$

The extracted results include the return loss S_{11} , radiation efficiency, total efficiency, gain, and two-dimensional and three-dimensional radiation patterns.

The CST geometry for the 10 mm separation case is shown in Figure 1.4.

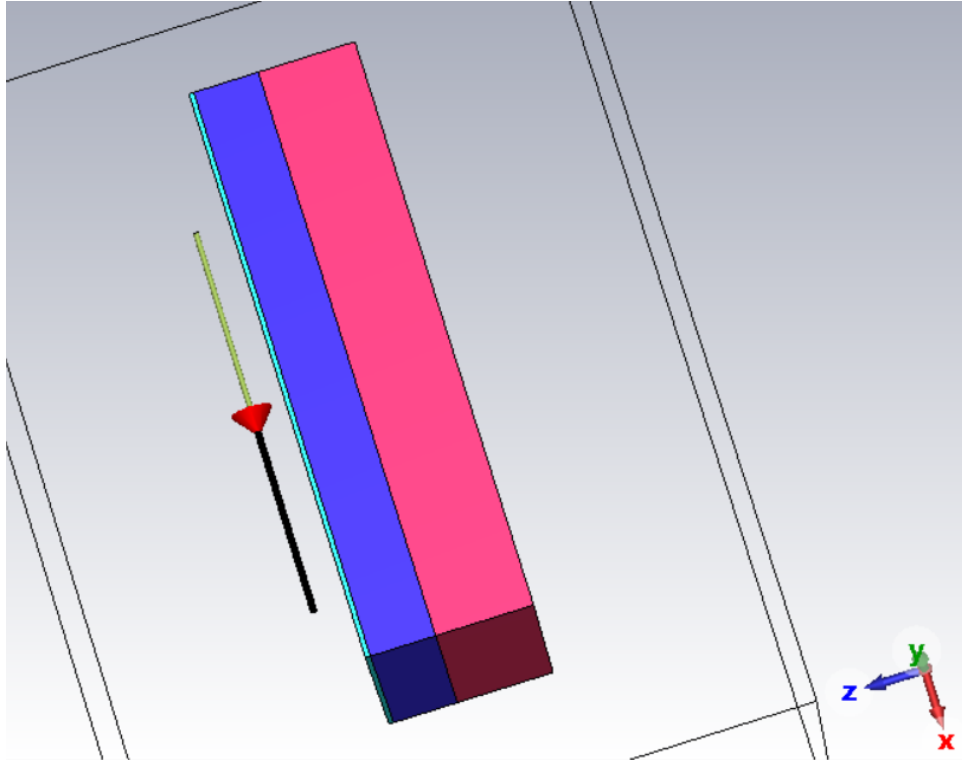


Figure 1.4: CST side-view geometry for the 10 mm antenna-body separation case.

Return-Loss Result

The CST return-loss result for the 10 mm body-separation case showed that the dominant matching minimum shifted below the target frequency. The minimum S_{11} occurred at

$$f_{\min} = 1.0312 \text{ GHz},$$

with

$$S_{11,\min} = -10.49934 \text{ dB}.$$

At the target frequency of 1.120 GHz, the return loss was

$$S_{11}(1.120 \text{ GHz}) = -7.839072 \text{ dB}.$$

Compared with the tuned free-space case, the antenna no longer satisfies the -10 dB matching criterion at the target frequency. The tuned free-space dipole had

Table 1.14: Return-loss result for the 10 mm body-separation case.

Quantity	Value
Dipole length	122.0 mm
Antenna-body separation	10 mm
Minimum- S_{11} frequency	1.0312 GHz
Minimum S_{11}	-10.49934 dB
S_{11} at 1.120 GHz	-7.839072 dB

$$S_{11}(1.120 \text{ GHz}) = -15.48433 \text{ dB},$$

whereas the 10 mm body case produced

$$S_{11}(1.120 \text{ GHz}) = -7.839072 \text{ dB}.$$

This degradation indicates that the lossy body model perturbs the antenna input impedance even at the largest separation distance considered in this project.

The return-loss response for the 10 mm body-loading case is shown in Figure 1.5.

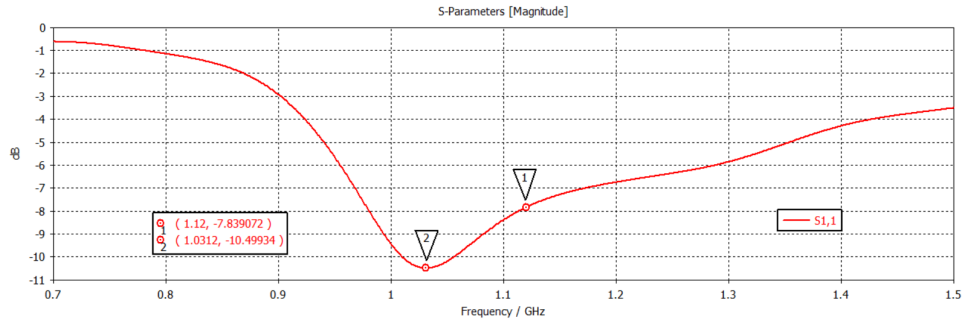


Figure 1.5: Return loss for the 10 mm body-separation case. The dominant matching minimum occurs at 1.0312 GHz, while S_{11} at 1.120 GHz is -7.839072 dB.

Efficiency and Gain Result

The far-field result at 1.120 GHz showed that the nearby body reduced the radiating performance of the antenna. CST reported the following values:

$$\eta_{\text{rad}} = -3.410 \text{ dB},$$

$$\eta_{\text{tot}} = -4.190 \text{ dB},$$

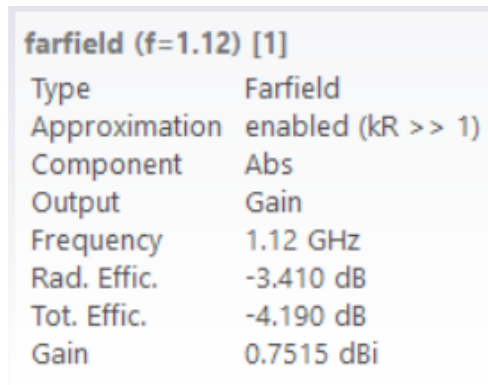
and

$$G = 0.7515 \text{ dBi.}$$

Table 1.15: Efficiency and gain result for the 10 mm body-separation case at 1.120 GHz.

Quantity	Value
Radiation efficiency	−3.410 dB
Total efficiency	−4.190 dB
Gain	0.7515 dBi

The radiation efficiency reduction is caused by power absorption in the lossy tissue layers. The total efficiency is lower than the radiation efficiency because it also includes impedance mismatch at the target frequency. Therefore, both tissue loss and mismatch contribute to the reduced antenna performance.



farfield (f=1.12) [1]	
Type	Farfield
Approximation	enabled (kR >> 1)
Component	Abs
Output	Gain
Frequency	1.12 GHz
Rad. Effic.	-3.410 dB
Tot. Effic.	-4.190 dB
Gain	0.7515 dBi

Figure 1.6: CST far-field parameter summary for the 10 mm body-separation case at 1.120 GHz.

Radiation Pattern Observation

The far-field monitor at 1.120 GHz was used to evaluate the radiation behavior of the complete antenna-body system. Although the body is located in the near-field region of the dipole, the CST far-field result represents the far-zone radiation pattern after the near-field interaction between the antenna and the body has been solved.

For the 10 mm case, the radiation pattern remained broadly dipole-like, but the presence of the lossy body model modified the field distribution and reduced the maximum gain. The body acts both as a high-permittivity loading structure and as a lossy absorber. As a result, part of the accepted power is dissipated in the tissue layers instead of being radiated into the far field.

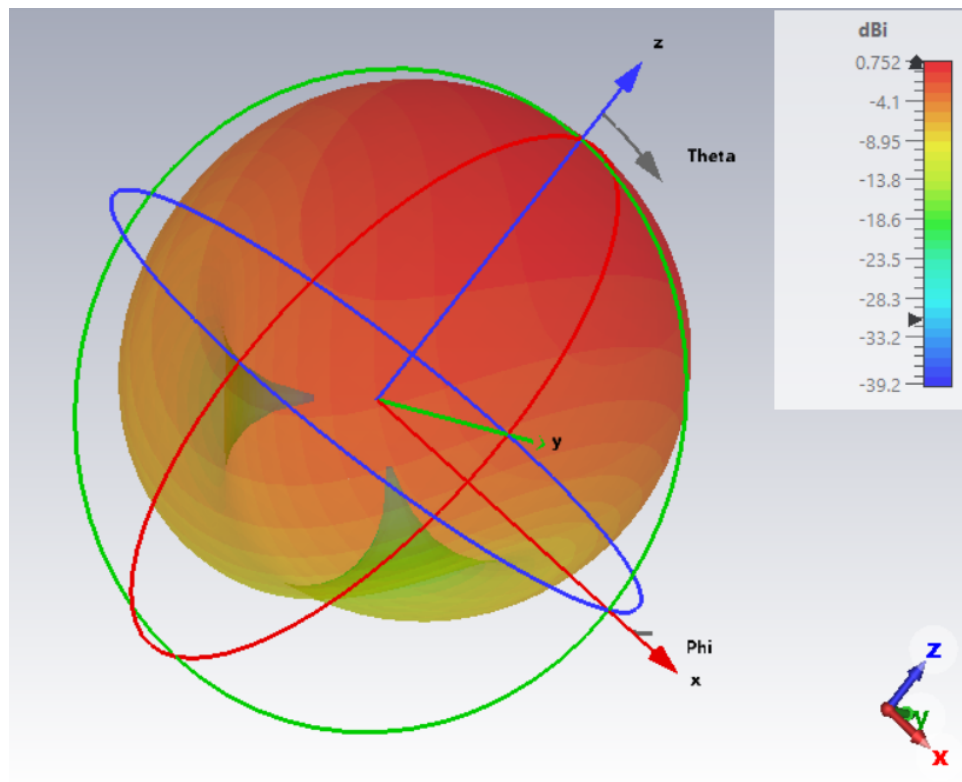


Figure 1.7: Three-dimensional gain pattern for the 10 mm body-separation case at 1.120 GHz.

The three-dimensional gain pattern at 1.120 GHz is shown in Figure 1.7.

The E-plane and H-plane radiation-pattern cuts for the 10 mm body-separation case are shown in Figures 1.8 and 1.9, respectively. Since the dipole is oriented along the x -axis, the $\phi = 0^\circ$ cut corresponds to the x - z principal plane, or E-plane, while the $\phi = 90^\circ$ cut corresponds to the y - z principal plane, or H-plane.

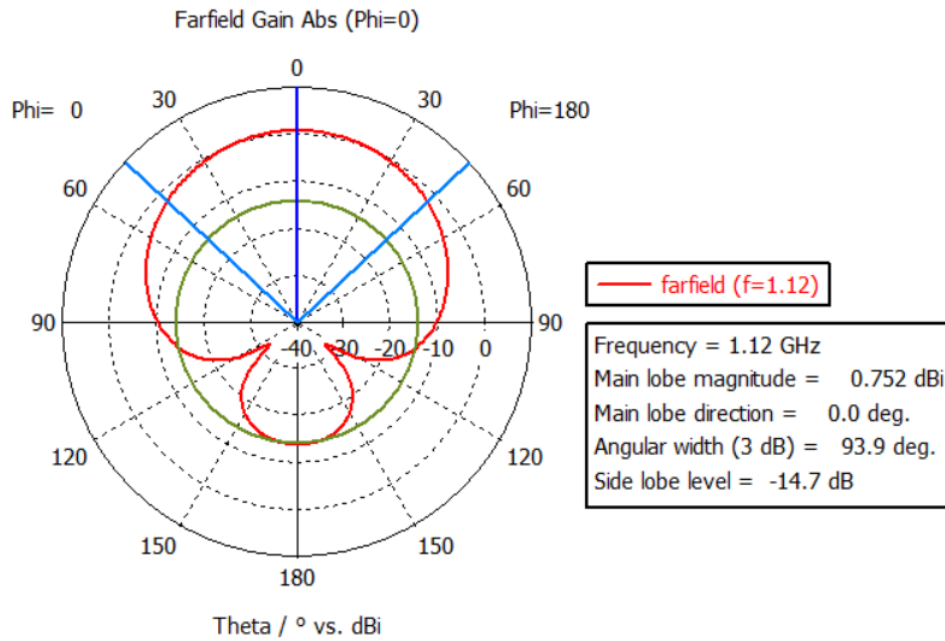


Figure 1.8: E-plane radiation-pattern cut for the 10 mm body-separation case at 1.120 GHz. This cut corresponds to $\phi = 0^\circ$, or the x - z principal plane.

Interpretation of the 10 mm Case

The 10 mm body-separation case shows the first clear evidence of body loading. Relative to the tuned free-space antenna, the matching minimum shifted from approximately

$$1.1168 \text{ GHz}$$

to

$$1.0312 \text{ GHz.}$$

This downward shift indicates that the nearby high-permittivity tissue layers increased

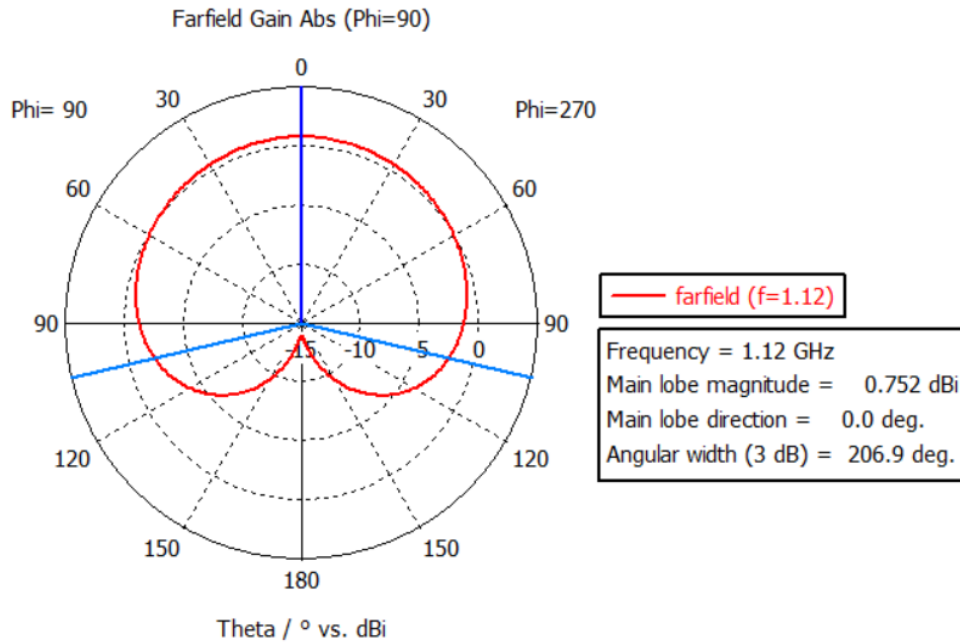


Figure 1.9: H-plane radiation-pattern cut for the 10 mm body-separation case at 1.120 GHz. This cut corresponds to $\phi = 90^\circ$, or the y - z principal plane.

the effective electrical loading of the dipole. In other words, the antenna behaves electrically longer when placed near the body.

At the target frequency, the return loss degraded from

$$-15.48433 \text{ dB}$$

in free space to

$$-7.839072 \text{ dB}$$

with the body at 10 mm. The gain was also reduced to

$$0.7515 \text{ dBi.}$$

These results confirm that even a 10 mm separation is close enough for the body to significantly affect the input matching and radiating performance of the wearable dipole antenna.

1.6.2 Body Separation of 5 mm

The 5 mm separation case represents a stronger body-loading condition than the 10 mm case. By reducing the air gap between the dipole and the multilayer body model, the antenna fields interact more strongly with the lossy tissue layers. Therefore, a larger impedance shift, stronger efficiency degradation, and greater radiation-pattern modification are expected.

Geometry and Simulation Setup

The same tuned free-space dipole was used for the 5 mm body-separation case. The total dipole length was

$$L = 122.0 \text{ mm},$$

the wire radius was

$$a = 0.905 \text{ mm},$$

and the center feed gap was

$$g = 2.0 \text{ mm}.$$

The separation distance was again defined as the air gap between the lower surface of the cylindrical dipole wire and the top surface of the skin layer. Since the lower surface of the wire is located at

$$z = -0.905 \text{ mm},$$

the top of the skin layer for the 5 mm case was placed at

$$z_{\text{skin,top}} = -(5 + 0.905) = -5.905 \text{ mm}.$$

The skin, fat, and muscle layers were stacked below this coordinate in the negative z -direction. The corresponding layer coordinates were summarized in Table 1.12. The tissue material properties, finite body patch dimensions, frequency range, solver settings, and boundary conditions were kept the same as in the 10 mm case.

The CST time-domain solver was used over the frequency range

$$0.7 \text{ GHz} \leq f \leq 1.5 \text{ GHz},$$

and the far-field monitor was defined at

$$f_0 = 1.120 \text{ GHz}.$$

The CST geometry for the 5 mm separation case is shown in Figure 1.10.

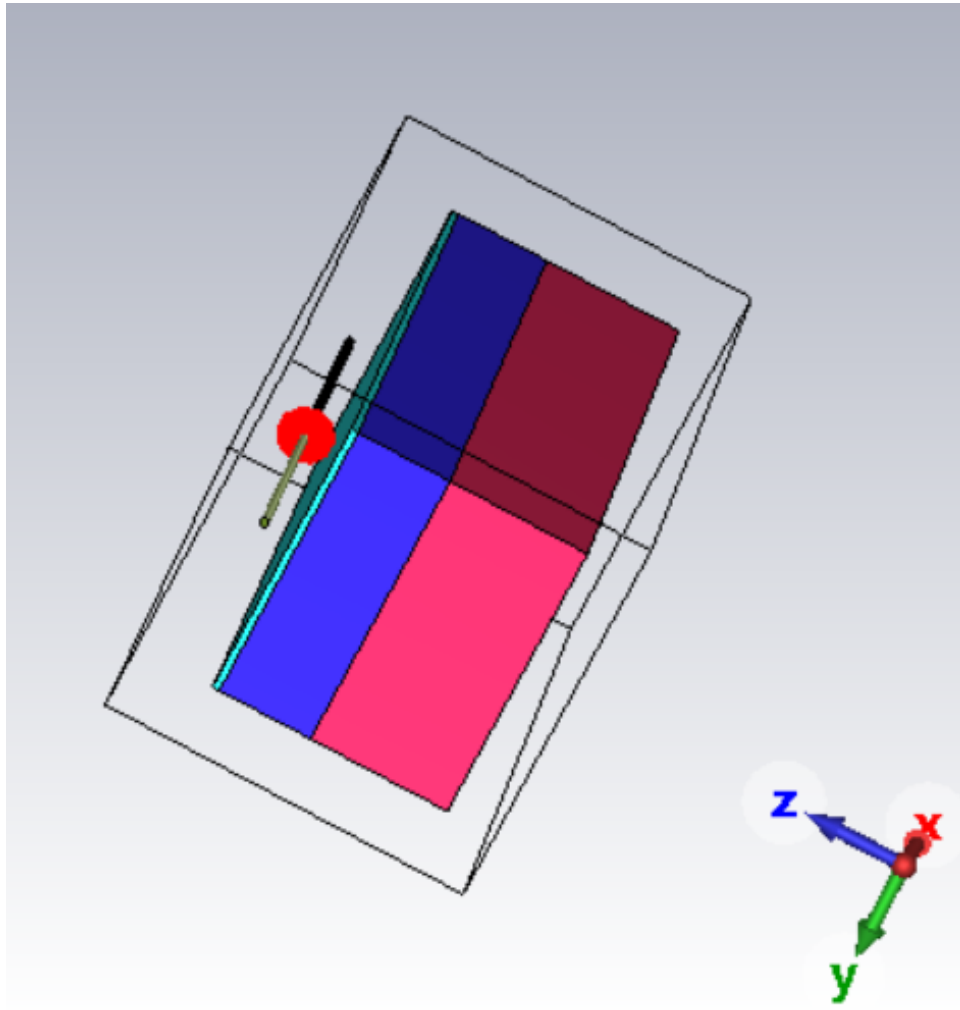


Figure 1.10: CST side-view geometry for the 5 mm antenna-body separation case.

Return-Loss Result

The CST return-loss result for the 5 mm body-separation case showed a stronger downward shift of the matching minimum compared with the 10 mm case. The minimum

S_{11} occurred at

$$f_{\min} = 0.9792 \text{ GHz},$$

with

$$S_{11,\min} = -11.9744 \text{ dB}.$$

At the design target frequency,

$$f_0 = 1.120 \text{ GHz},$$

the return loss was

$$S_{11}(1.120 \text{ GHz}) = -5.903831 \text{ dB}.$$

Table 1.16: Return-loss result for the 5 mm body-separation case.

Quantity	Value
Dipole length	122.0 mm
Antenna-body separation	5 mm
Minimum- S_{11} frequency	0.9792 GHz
Minimum S_{11}	-11.9744 dB
S_{11} at 1.120 GHz	-5.903831 dB

Compared with the 10 mm case, the matching minimum moved from

$$1.0312 \text{ GHz}$$

to

$$0.9792 \text{ GHz}.$$

This additional downward shift confirms that reducing the antenna-body separation increases the effective dielectric loading of the dipole. At the target frequency, the return loss also degraded from

$$-7.839072 \text{ dB}$$

for the 10 mm case to

$$-5.903831 \text{ dB}$$

for the 5 mm case.

The return-loss response for the 5 mm body-loading case is shown in Figure 1.11.

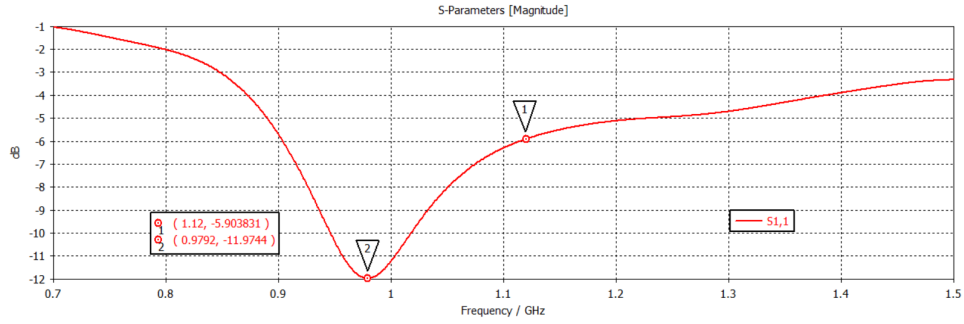


Figure 1.11: Return loss for the 5 mm body-separation case. The matching minimum occurs at 0.9792 GHz, while S_{11} at 1.120 GHz is -5.903831 dB.

Efficiency and Gain Result

The far-field result at 1.120 GHz showed that the 5 mm body-separation case had lower radiating performance than the 10 mm case. CST reported the following values:

$$\eta_{\text{rad}} = -3.837 \text{ dB},$$

$$\eta_{\text{tot}} = -5.126 \text{ dB},$$

and

$$G = 0.3534 \text{ dBi}.$$

Table 1.17: Efficiency and gain result for the 5 mm body-separation case at 1.120 GHz.

Quantity	Value
Radiation efficiency	-3.837 dB
Total efficiency	-5.126 dB
Gain	0.3534 dBi

The radiation efficiency decreased because the tissue layers were closer to the antenna and absorbed more electromagnetic power. The total efficiency decreased even more because the antenna was also poorly matched at 1.120 GHz. Therefore, the lower total efficiency is the combined result of increased tissue loss and increased impedance mismatch.

The CST far-field parameter summary for the 5 mm body-separation case is shown in Figure 1.12.

farfield (f=1.12) [1]	
Type	Farfield
Approximation	enabled (kR >> 1)
Component	Abs
Output	Gain
Frequency	1.12 GHz
Rad. Effic.	-3.837 dB
Tot. Effic.	-5.126 dB
Gain	0.3534 dBi

Figure 1.12: CST far-field parameter summary for the 5 mm body-separation case at 1.120 GHz.

Radiation Pattern Observation

The far-field radiation pattern was evaluated at

1.120 GHz.

As in the 10 mm case, the CST far-field monitor reports the far-zone radiation behavior of the complete antenna-body system. The body itself is located in the near-field region of the antenna, but the far-field pattern is obtained after solving the near-field interaction between the dipole and the lossy tissue layers.

For the 5 mm case, the radiation pattern was more strongly affected by the body than in the 10 mm case. The maximum gain decreased to

0.3534 dBi,

indicating that less power was radiated effectively into the far field. This reduction is consistent with stronger tissue absorption and stronger perturbation of the current distribution on the dipole.

The three-dimensional gain pattern at 1.120 GHz is shown in Figure 1.13.

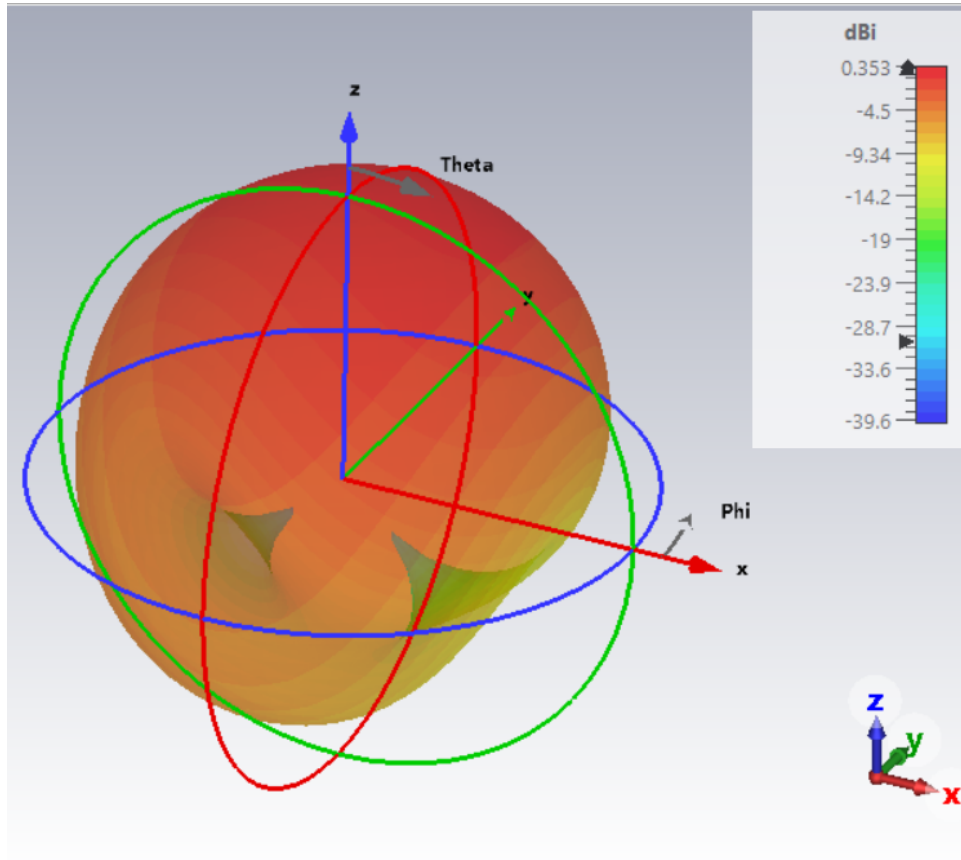


Figure 1.13: Three-dimensional gain pattern for the 5 mm body-separation case at 1.120 GHz.

The E-plane and H-plane radiation-pattern cuts for the 5 mm body-separation case are shown in Figures 1.14 and 1.15, respectively. Since the dipole is oriented along the x -axis, the $\phi = 0^\circ$ cut corresponds to the x - z principal plane, or E-plane, while the $\phi = 90^\circ$ cut corresponds to the y - z principal plane, or H-plane.

Interpretation of the 5 mm Case

The 5 mm body-separation case confirms that antenna-body coupling becomes stronger as the antenna is moved closer to the lossy multilayer tissue model. The tuned free-space dipole had its matching minimum near

$$1.1168 \text{ GHz},$$

whereas the 5 mm body case produced a matching minimum at

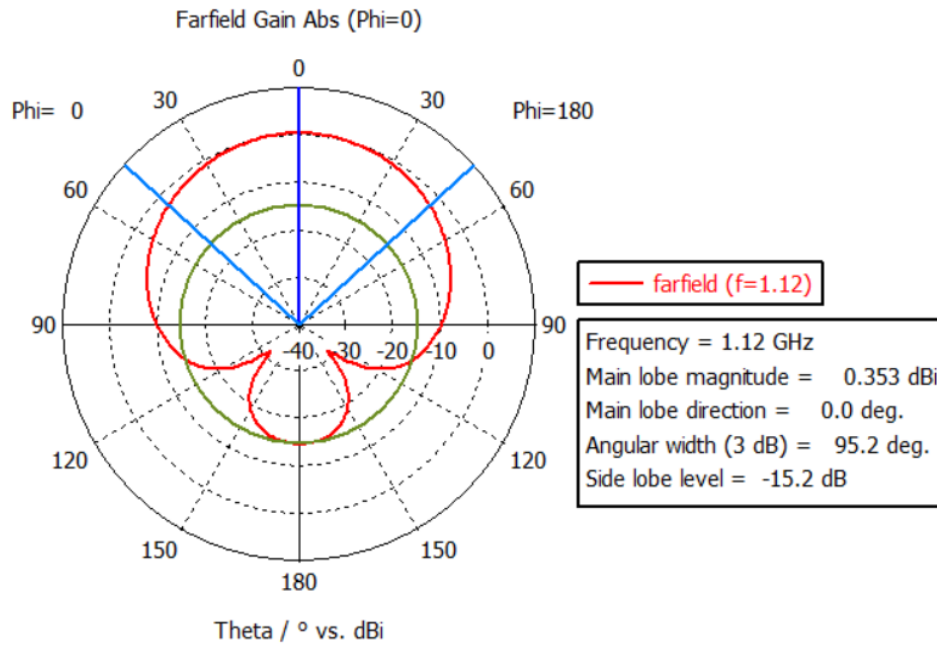


Figure 1.14: E-plane radiation-pattern cut for the 5 mm body-separation case at 1.120 GHz. For the x -oriented dipole, this cut corresponds to the $\phi = 0^\circ$ principal plane, or the x - z plane.

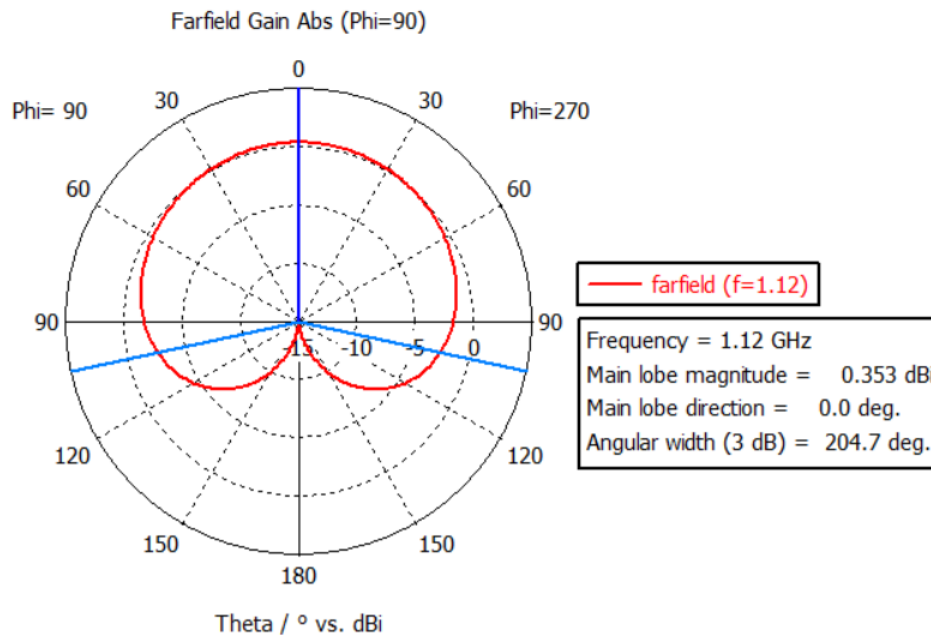


Figure 1.15: H-plane radiation-pattern cut for the 5 mm body-separation case at 1.120 GHz. For the x -oriented dipole, this cut corresponds to the $\phi = 90^\circ$ principal plane, or the y - z plane.

0.9792 GHz.

This shift corresponds to a significant detuning below the design operating frequency. The physical reason is that the high-permittivity body layers increase the effective electrical loading around the dipole, causing the fixed physical antenna length to behave as if it were electrically longer.

At the target frequency, the return loss degraded to

-5.903831 dB,

which is far above the desired -10 dB matching criterion. The gain also dropped to

0.3534 dBi.

Therefore, the 5 mm case demonstrates both impedance degradation and radiating-performance degradation. This case was later selected as the main retuning case because it represents a realistic near-body wearable condition and shows a clear downward detuning trend that can be addressed by shortening the dipole.

1.6.3 Body Separation of 0.1 mm

The 0.1 mm separation case represents the strongest near-field loading condition in this study. At this extremely small spacing, the lossy multilayer tissue model is almost in contact with the dipole. Therefore, the antenna response should be interpreted as a strongly coupled antenna-body problem rather than as a lightly perturbed half-wave dipole.

Geometry and Simulation Setup

The same tuned free-space dipole geometry was used for the 0.1 mm body-separation case. The dipole length was

$L = 122.0$ mm,

the wire radius was

$a = 0.905$ mm,

and the center feed gap was

$$g = 2.0 \text{ mm.}$$

The separation distance was defined as the air gap between the lower surface of the cylindrical dipole wire and the top surface of the skin layer. Since the lower surface of the wire was located at

$$z = -0.905 \text{ mm,}$$

the top of the skin layer for the 0.1 mm case was placed at

$$z_{\text{skin,top}} = -(0.1 + 0.905) = -1.005 \text{ mm.}$$

The skin, fat, and muscle layers were stacked below this coordinate in the negative z -direction. The corresponding layer coordinates were summarized in Table 1.13. The material properties, finite body patch dimensions, solver settings, frequency range, and boundary settings were kept consistent with the previous body-separation cases.

The CST time-domain solver was used over the frequency range

$$0.7 \text{ GHz} \leq f \leq 1.5 \text{ GHz,}$$

and the far-field monitor was defined at the design target frequency,

$$f_0 = 1.120 \text{ GHz.}$$

The CST side-view geometry for the extreme 0.1 mm antenna-body separation case is shown in Figure 1.16.

Return-Loss Result

The 0.1 mm separation case produced a different return-loss behavior compared with the 10 mm and 5 mm cases. Instead of a simple downward shift of the matching minimum, the dominant S_{11} minimum appeared at a higher frequency. The CST result showed that the dominant matching minimum occurred at

$$f_{\text{min}} = 1.3976 \text{ GHz,}$$

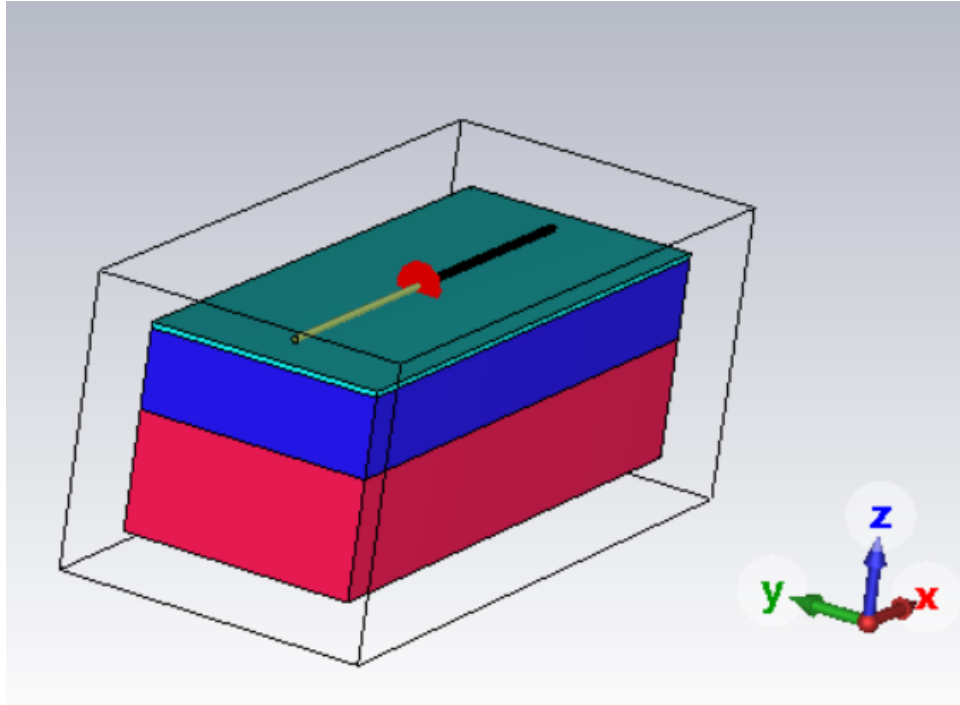


Figure 1.16: CST side-view geometry for the 0.1 mm antenna-body separation case.

with

$$S_{11,\min} = -13.8654 \text{ dB.}$$

At the design target frequency,

$$f_0 = 1.120 \text{ GHz,}$$

the return loss was

$$S_{11}(1.120 \text{ GHz}) = -6.951098 \text{ dB.}$$

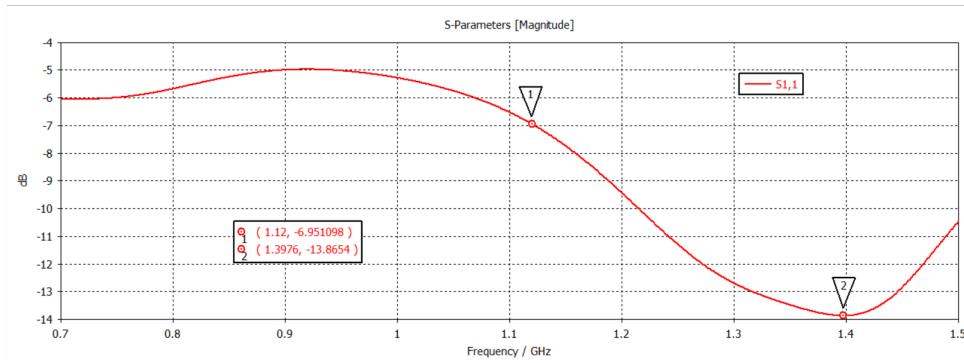
The higher-frequency dominant matching minimum should not be interpreted as a simple free-space half-wave resonance. At this extremely small separation distance, the antenna is strongly loaded by the nearby lossy and high-permittivity tissue layers. The input impedance is significantly perturbed by reactive near-field coupling, dielectric loading, and tissue loss. Therefore, the observed minimum represents the matching behavior of the coupled antenna-body system.

The return-loss response for the extreme 0.1 mm body-separation case is shown in

Table 1.18: Return-loss result for the 0.1 mm body-separation case.

Quantity	Value
Dipole length	122.0 mm
Antenna-body separation	0.1 mm
Dominant matching-minimum frequency	1.3976 GHz
Minimum S_{11}	−13.8654 dB
S_{11} at 1.120 GHz	−6.951098 dB

Figure 1.17.

**Figure 1.17:** Return loss for the 0.1 mm body-separation case. The dominant matching minimum occurs at 1.3976 GHz, while S_{11} at 1.120 GHz is −6.951098 dB.

Efficiency and Gain Result

The far-field result at 1.120 GHz showed a significant reduction in antenna performance for the 0.1 mm case. CST reported the following values:

$$\eta_{\text{rad}} = -10.52 \text{ dB},$$

$$\eta_{\text{tot}} = -11.50 \text{ dB},$$

and

$$G = -5.261 \text{ dBi}.$$

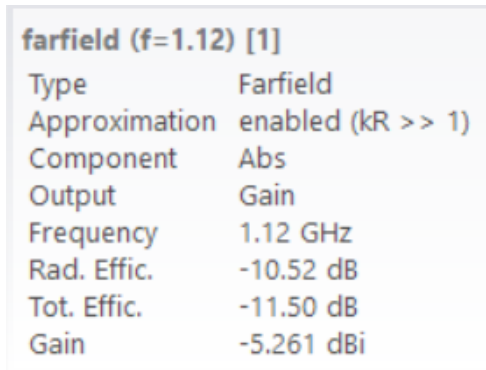
This result shows that the extremely small air gap causes substantial degradation in radiating performance. The radiation efficiency is much lower than in the 10 mm and 5 mm cases because a larger portion of the accepted power is absorbed in the lossy

Table 1.19: Efficiency and gain result for the 0.1 mm body-separation case at 1.120 GHz.

Quantity	Value
Radiation efficiency	−10.52 dB
Total efficiency	−11.50 dB
Gain	−5.261 dBi

tissue layers. The negative gain value indicates that the antenna-body system radiates much less effectively at the target frequency than the free-space tuned dipole.

The CST far-field parameter summary for the extreme 0.1 mm body-separation case is shown in Figure 1.18.



farfield (f=1.12) [1]	
Type	Farfield
Approximation	enabled (kR >> 1)
Component	Abs
Output	Gain
Frequency	1.12 GHz
Rad. Effic.	-10.52 dB
Tot. Effic.	-11.50 dB
Gain	-5.261 dBi

Figure 1.18: CST far-field parameter summary for the 0.1 mm body-separation case at 1.120 GHz.

Radiation Pattern Observation

The far-field radiation pattern for the 0.1 mm case was evaluated at

1.120 GHz.

Although the body is located in the near-field region of the antenna, the CST far-field monitor computes the far-zone radiation behavior of the complete antenna-body structure. Therefore, the reported pattern represents the radiated field after the strong near-field coupling and tissue absorption have already affected the antenna current distribution and accepted power.

For the 0.1 mm separation case, the radiation pattern is expected to be more strongly distorted than in the 10 mm and 5 mm cases. The body is located extremely close to the dipole, so the lossy tissue layers strongly perturb the local electromagnetic fields. The maximum gain at the target frequency was

$$G = -5.261 \text{ dBi},$$

which confirms a severe reduction in radiated performance. The three-dimensional gain pattern for the extreme 0.1 mm body-separation case is shown in Figure 1.19.

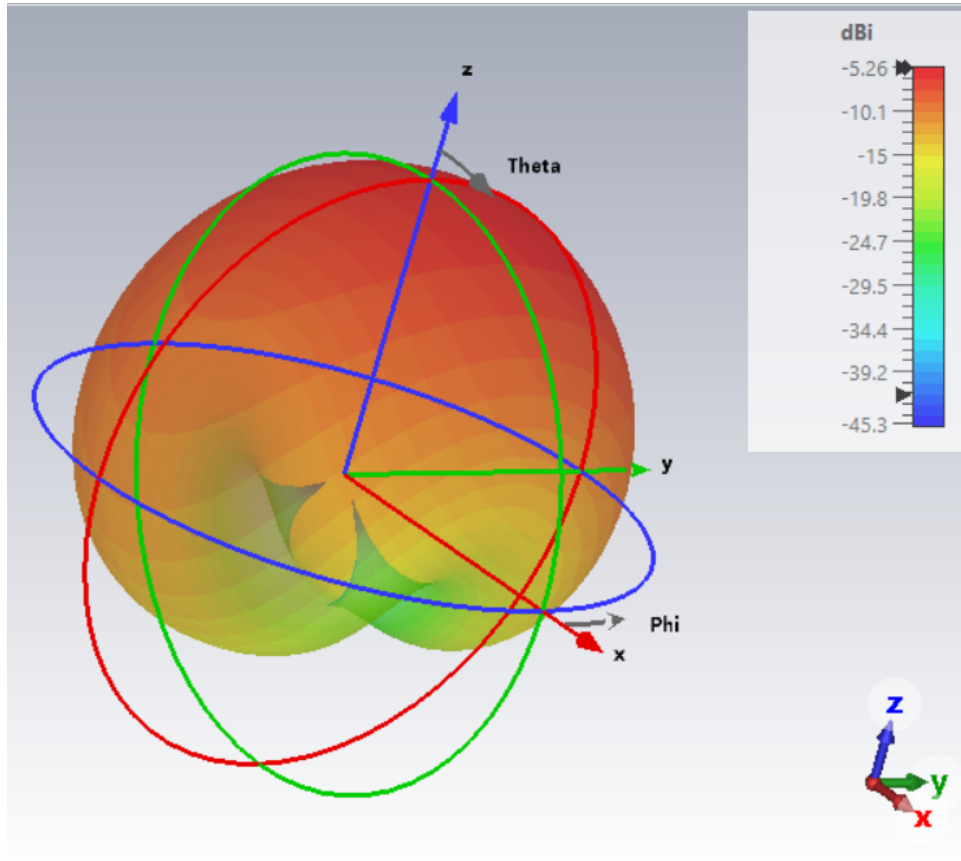


Figure 1.19: Three-dimensional gain pattern for the 0.1 mm body-separation case at 1.120 GHz.

The E-plane and H-plane radiation-pattern cuts for the extreme 0.1 mm body-separation case are shown in Figures 1.20 and 1.21, respectively. Since the dipole is oriented along the x -axis, the $\phi = 0^\circ$ cut corresponds to the x - z principal plane, or E-plane, while the $\phi = 90^\circ$ cut corresponds to the y - z principal plane, or H-plane.

Near-Field Interpretation of the 0.1 mm Case

The 0.1 mm case requires special interpretation because the antenna-body distance is extremely small compared with the free-space wavelength. At

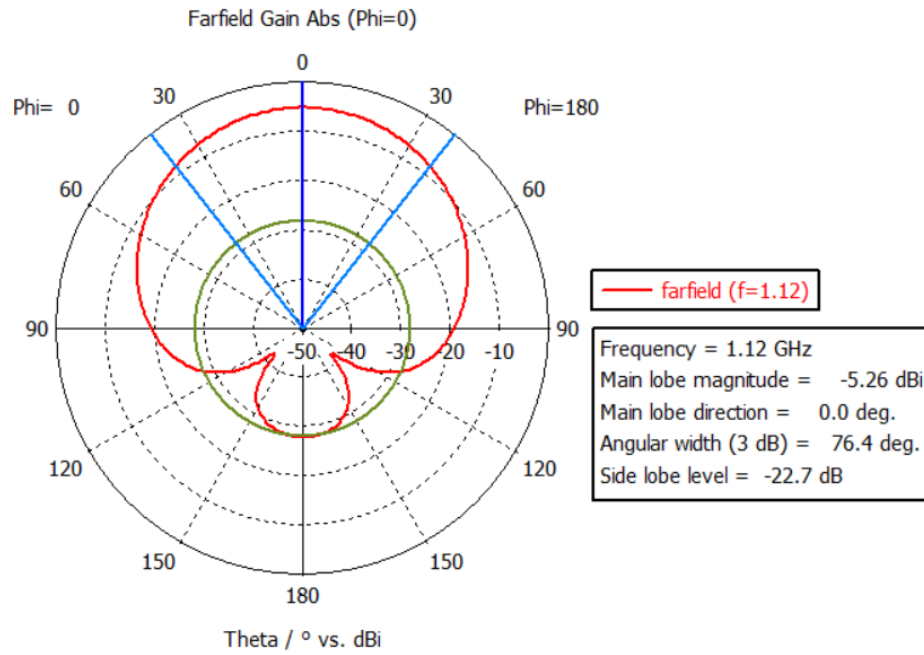


Figure 1.20: E-plane radiation-pattern cut for the 0.1 mm body-separation case at 1.120 GHz. This cut corresponds to $\phi = 0^\circ$, or the x - z principal plane.

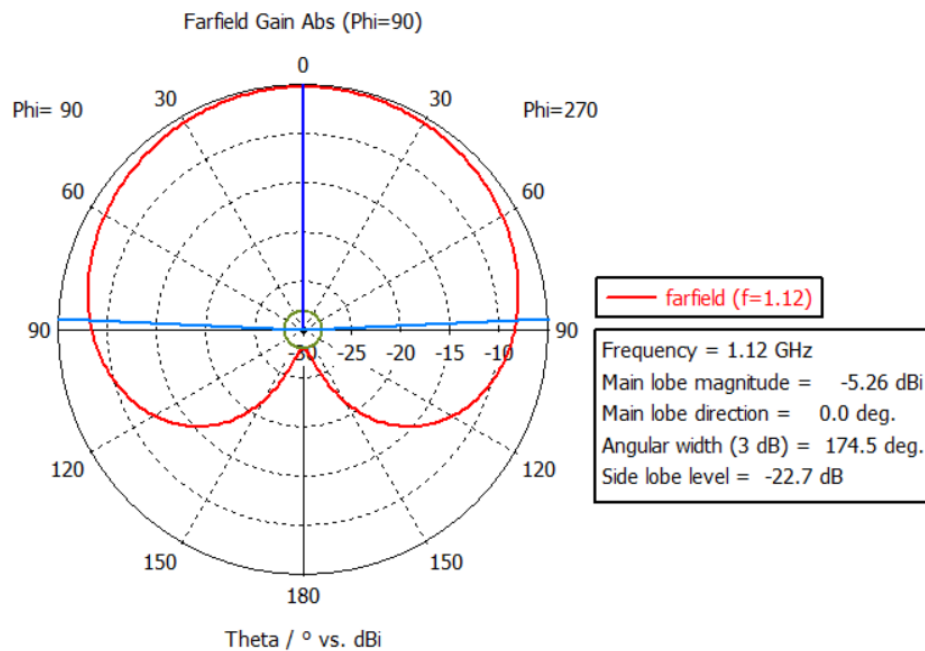


Figure 1.21: H-plane radiation-pattern cut for the 0.1 mm body-separation case at 1.120 GHz. This cut corresponds to $\phi = 90^\circ$, or the y - z principal plane.

$$f_0 = 1.120 \text{ GHz},$$

the wavelength is approximately

$$\lambda_0 = 267.86 \text{ mm}.$$

Therefore,

$$0.1 \text{ mm} \approx 0.00037\lambda_0.$$

This means that the body is deeply inside the reactive near-field region of the antenna. In this region, the electric and magnetic fields contain strong stored-energy components. When a high-permittivity and lossy material is placed in this region, it can substantially modify the antenna input impedance and current distribution.

For the 10 mm and 5 mm cases, the dominant matching minimum shifted downward in frequency, which is consistent with increased effective dielectric loading. However, for the 0.1 mm case, the coupling is so strong that the return-loss response no longer follows the same monotonic trend. The dominant matching minimum appeared at

$$1.3976 \text{ GHz},$$

while the return loss at the target frequency remained poor:

$$S_{11}(1.120 \text{ GHz}) = -6.951098 \text{ dB}.$$

This behavior indicates that the antenna is no longer behaving as a lightly perturbed free-space dipole. Instead, it behaves as a strongly coupled lossy antenna-body system. The higher-frequency matching minimum may result from a different impedance condition created by the combined effects of reactive loading, dielectric loss, finite body geometry, and strong local field interaction.

Interpretation of the 0.1 mm Case

Numerical-reliability caveat. The upward displacement of the dominant matching minimum in this extreme-clearance case is non-monotonic relative to the 10 mm and 5 mm cases. Because an independent mesh-convergence and boundary-convergence study could not be completed within the CST Learning Edition cell limit, the 1.3976 GHz

minimum is retained as an observed result of the documented finite model, not as a convergence-verified physical trend. The target-frequency mismatch and severe efficiency degradation are the more robust comparative observations for this case.

The 0.1 mm body-separation case produced the most severe performance degradation among the three body-loading cases. Although the dominant S_{11} minimum was below -10 dB, it occurred far above the target frequency. At the design operating frequency of 1.120 GHz, the antenna remained poorly matched.

The most important result of this case is the strong reduction in radiation efficiency and gain. The radiation efficiency decreased to

$$-10.52 \text{ dB},$$

and the gain decreased to

$$-5.261 \text{ dBi}.$$

This confirms that placing the dipole almost in contact with the lossy body model causes significant power absorption and severe degradation of radiated performance. Therefore, the 0.1 mm case represents an extreme near-body loading condition and illustrates why wearable antennas usually require careful spacing, insulation, matching, or redesign when operating close to the human body.

1.7 Redesign and Retuning

The body-loaded simulations showed that the free-space-tuned dipole no longer remained well matched at the design operating frequency. Since the 5 mm case exhibited a clear downward shift of the matching minimum, it was selected for a first-order retuning study based on shortening the dipole length.

1.7.1 Motivation for Retuning

The tuned free-space dipole had good impedance matching at the target frequency,

$$f_0 = 1.120 \text{ GHz},$$

with

$$S_{11}(1.120 \text{ GHz}) = -15.48433 \text{ dB}.$$

However, when the same dipole was placed 5 mm above the body model, the matching minimum shifted downward to

$$f_{\min} = 0.9792 \text{ GHz},$$

and the return loss at the target frequency degraded to

$$S_{11}(1.120 \text{ GHz}) = -5.903831 \text{ dB}.$$

This indicates that the 122.0 mm dipole became electrically too long in the presence of the nearby high-permittivity body model. Therefore, the first retuning strategy was to shorten the dipole length.

1.7.2 First-Order Length Estimate

A first-order retuning estimate was obtained by using the approximate inverse relationship between resonant frequency and dipole length,

$$f_r \propto \frac{1}{L}.$$

The estimated new length was calculated as

$$L_{\text{new}} = L_{\text{old}} \frac{f_r}{f_0},$$

where L_{old} is the original tuned free-space dipole length, f_r is the observed matching-minimum frequency near the body, and f_0 is the desired operating frequency.

For the 5 mm body-separation case,

$$L_{\text{old}} = 122.0 \text{ mm},$$

$$f_r = 0.9792 \text{ GHz},$$

and

$$f_0 = 1.120 \text{ GHz.}$$

Therefore,

$$L_{\text{new}} = 122.0 \frac{0.9792}{1.120} = 106.66 \text{ mm.}$$

The Python verification of this body-loaded retuning estimate is included in Appendix A, Listing A.1.

Based on this first-order estimate, the retuned CST dipole length was selected as

$$L = 106.6 \text{ mm.}$$

With the 2 mm center feed gap, the corresponding arm length was

$$L_{\text{arm}} = \frac{106.6 - 2.0}{2} = 52.3 \text{ mm.}$$

Thus, the retuned CST geometry used the right and left arm coordinates listed in Table 1.20.

Table 1.20: Retuned dipole geometry for $L = 106.6$ mm and 2 mm feed gap.

Parameter	Value
Total dipole length	106.6 mm
Feed gap	2.0 mm
Wire radius	0.905 mm
Each arm length	52.3 mm
Right arm x -range	1.0 mm to 53.3 mm
Left arm x -range	−53.3 mm to −1.0 mm
Port reference impedance	50 Ω

1.7.3 Retuned Result for $L = 106.6$ mm

The $L = 106.6$ mm retuned dipole was simulated above the same 5 mm body model. The CST result showed that the matching minimum moved closer to the target frequency. The minimum occurred at

$$f_{\text{min}} = 1.0928 \text{ GHz,}$$

with

$$S_{11,\min} = -8.152857 \text{ dB.}$$

At the target frequency,

$$f_0 = 1.120 \text{ GHz,}$$

the return loss improved to

$$S_{11}(1.120 \text{ GHz}) = -8.000936 \text{ dB.}$$

Table 1.21: Retuned result for the 5 mm body-separation case using $L = 106.6$ mm.

Quantity	Value
Antenna-body separation	5 mm
Retuned dipole length	106.6 mm
Feed gap	2.0 mm
Minimum- S_{11} frequency	1.0928 GHz
Minimum S_{11}	-8.152857 dB
S_{11} at 1.120 GHz	-8.000936 dB

Compared with the original 5 mm body case, the return loss at the target frequency improved from

$$-5.903831 \text{ dB}$$

to

$$-8.000936 \text{ dB.}$$

This confirms that shortening the antenna partially compensated for the downward detuning caused by the body model.

The return-loss response after shortening the dipole to 106.6 mm for the 5 mm body-separation case is shown in Figure 1.22.

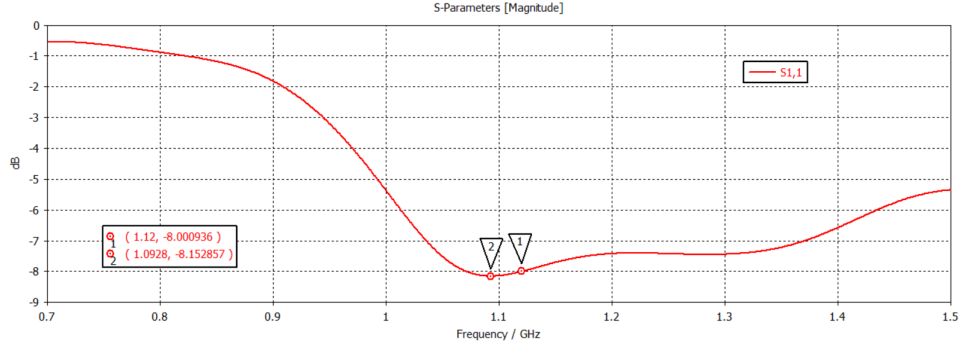


Figure 1.22: Return loss for the 5 mm body-separation case after retuning the dipole length to 106.6 mm.

1.7.4 Additional Length-Tuning Iterations

Additional length-tuning simulations were performed to determine whether further shortening could improve the match at 1.120 GHz. Two additional lengths were tested:

$$L = 105.8 \text{ mm}$$

and

$$L = 104.0 \text{ mm}.$$

For the $L = 105.8$ mm case, CST showed

$$f_{\min} = 1.1008 \text{ GHz},$$

$$S_{11,\min} = -7.970294 \text{ dB},$$

and

$$S_{11}(1.120 \text{ GHz}) = -7.904127 \text{ dB}.$$

For the $L = 104.0$ mm case, the dominant matching minimum appeared at

$$f_{\min} = 1.3208 \text{ GHz},$$

with

$$S_{11,\min} = -7.866346 \text{ dB.}$$

At the target frequency,

$$S_{11}(1.120 \text{ GHz}) = -7.569315 \text{ dB.}$$

These additional simulations did not improve the return loss at the target frequency compared with the $L = 106.6$ mm case.

Table 1.22: Length-tuning iterations for the 5 mm body-separation case.

Case	L (mm)	Gap (mm)	$f_{\min}$ (GHz)	$S_{11}(1.120 \text{ GHz})$ (dB)
Original body 5 mm	122.0	2.0	0.9792	-5.903831
Retuned L106.6	106.6	2.0	1.0928	-8.000936
Retuned L105.8	105.8	2.0	1.1008	-7.904127
Retuned L104.0	104.0	2.0	1.3208	-7.569315

The best length-only retuning result among the tested cases was obtained with

$$L = 106.6 \text{ mm.}$$

Although this case did not fully satisfy the -10 dB matching criterion, it provided the best target-frequency return loss among the length-only retuned designs.

The return-loss response for the additional $L = 105.8$ mm tuning case is shown in Figure 1.23.

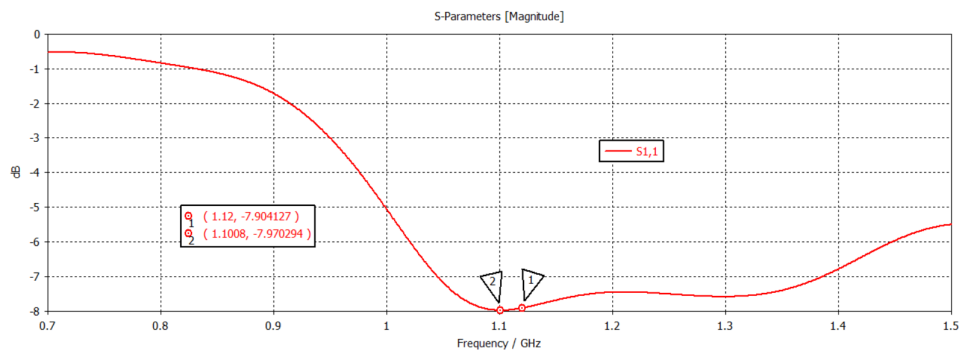


Figure 1.23: Return loss for the 5 mm body-separation case with retuned dipole length $L = 105.8$ mm.

The return-loss response for the shorter $L = 104.0$ mm tuning case is shown in Figure 1.24.

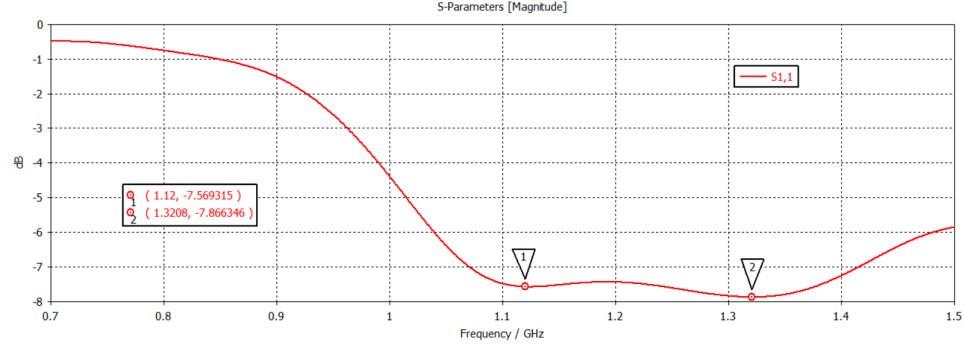


Figure 1.24: Return loss for the 5 mm body-separation case with retuned dipole length $L = 104.0$ mm.

1.7.5 Feed-Gap Variation

A feed-gap variation was also tested to examine whether changing the feed geometry could improve the impedance match. The dipole length was kept at

$$L = 106.6 \text{ mm},$$

but the feed gap was reduced from

$$2.0 \text{ mm}$$

to

$$1.0 \text{ mm}.$$

With a 1.0 mm feed gap, each arm length became

$$L_{\text{arm}} = \frac{106.6 - 1.0}{2} = 52.8 \text{ mm}.$$

The corresponding CST geometry is summarized in Table 1.23.

The CST result for the 1 mm gap case was

Table 1.23: Retuned dipole geometry for $L = 106.6$ mm and 1 mm feed gap.

Parameter	Value
Total dipole length	106.6 mm
Feed gap	1.0 mm
Wire radius	0.905 mm
Each arm length	52.8 mm
Right arm x -range	0.5 mm to 53.3 mm
Left arm x -range	−53.3 mm to −0.5 mm
Port location	$x = -0.5$ mm to $x = 0.5$ mm
Port reference impedance	50 Ω

$$f_{\min} = 1.0968 \text{ GHz},$$

$$S_{11,\min} = -7.925415 \text{ dB},$$

and

$$S_{11}(1.120 \text{ GHz}) = -7.821004 \text{ dB}.$$

The far-field result at 1.120 GHz was

$$\eta_{\text{rad}} = -3.899 \text{ dB},$$

$$\eta_{\text{tot}} = -4.683 \text{ dB},$$

and

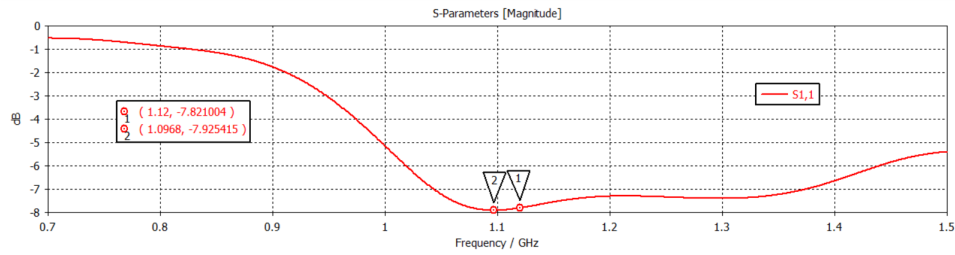
$$G = 0.1812 \text{ dBi}.$$

Although reducing the feed gap slightly moved the matching minimum, it did not improve the return loss at the target frequency compared with the $L = 106.6$ mm, 2 mm gap case. Therefore, the 1 mm feed-gap case was not selected as the best retuned configuration.

The return-loss response for the $L = 106.6$ mm, 1 mm feed-gap tuning case is shown in Figure 1.25.

Table 1.24: Feed-gap variation result for the 5 mm body-separation case.

Quantity	Value
Dipole length	106.6 mm
Feed gap	1.0 mm
Minimum- S_{11} frequency	1.0968 GHz
Minimum S_{11}	-7.925415 dB
S_{11} at 1.120 GHz	-7.821004 dB
Radiation efficiency at 1.120 GHz	-3.899 dB
Total efficiency at 1.120 GHz	-4.683 dB
Gain at 1.120 GHz	0.1812 dBi

**Figure 1.25:** Return loss for the 5 mm body-separation case with $L = 106.6$ mm and 1 mm feed gap.

The CST far-field parameter summary for the $L = 106.6$ mm, 1 mm feed-gap tuning case is shown in Figure 1.26.

farfield (f=1.12) [1]	
Type	Farfield
Approximation	enabled (kR >> 1)
Component	Abs
Output	Gain
Frequency	1.12 GHz
Rad. Effic.	-3.899 dB
Tot. Effic.	-4.683 dB
Gain	0.1812 dBi

Figure 1.26: CST far-field parameter summary for the 5 mm body-separation case with $L = 106.6$ mm and 1 mm feed gap at 1.120 GHz.

The three-dimensional gain pattern for the $L = 106.6$ mm, 1 mm feed-gap tuning case is shown in Figure 1.27.

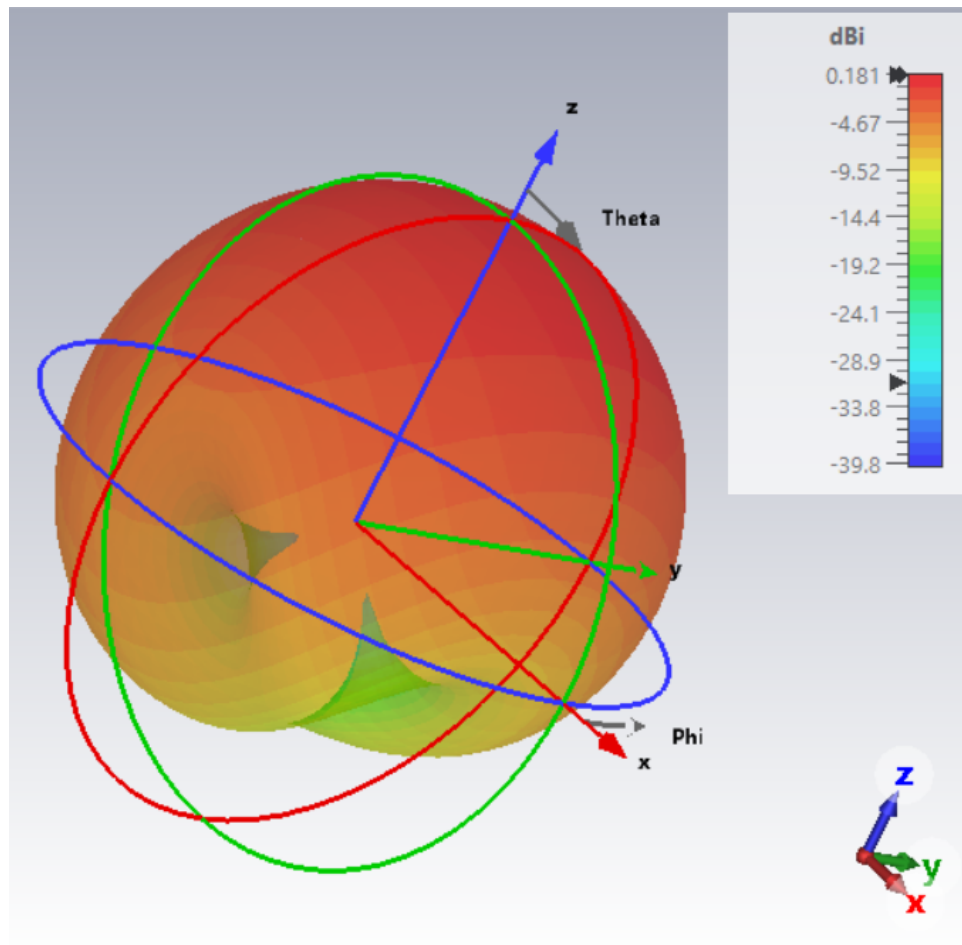


Figure 1.27: Three-dimensional gain pattern for the $L = 106.6$ mm, 1 mm feed-gap case at 1.120 GHz.

1.7.6 Retuning Summary

The retuning results are summarized in Table 1.25. The original 5 mm body case was poorly matched at the target frequency. Length-only retuning improved the target-frequency return loss, but it did not fully restore the -10 dB matching condition.

Table 1.25: Summary of retuning results for the 5 mm body-separation case.

Case	L (mm)	Gap (mm)	$f_{\min}$ (GHz)	$S_{11}(1.120 \text{ GHz})$ (dB)
Original body 5 mm	122.0	2.0	0.9792	-5.903831
Retuned L106.6	106.6	2.0	1.0928	-8.000936
Retuned L105.8	105.8	2.0	1.1008	-7.904127
Retuned L104.0	104.0	2.0	1.3208	-7.569315
Retuned L106.6 Gap1	106.6	1.0	1.0968	-7.821004

The best tested retuned configuration was

$$L = 106.6 \text{ mm},$$

with a

$$2.0 \text{ mm}$$

feed gap. This configuration improved the return loss at the target frequency from

$$-5.903831 \text{ dB}$$

to

$$-8.000936 \text{ dB}.$$

However, the result remained above the desired -10 dB threshold. This indicates that length-only retuning is not sufficient for complete impedance recovery near the lossy body model.

A more complete wearable antenna redesign would require additional techniques, such as adjusting the antenna height, modifying the conductor radius, changing the feed structure, using an impedance-matching network, adding an insulating spacer, or using a different wearable antenna geometry such as a planar or backed structure.

1.8 Engineering Discussion

The numerical results are most useful when they are interpreted together rather than as isolated CST outputs. The following discussion connects the observed S_{11} shifts, efficiency reduction, gain degradation, radiation-pattern changes, and retuning results to the underlying electromagnetic behavior of a wearable antenna near a lossy body model.

1.8.1 Overall Return-Loss Comparison

The return-loss results show that the multilayer body model significantly changes the antenna input behavior. The tuned free-space dipole was well matched near the design target frequency of

$$f_0 = 1.120 \text{ GHz.}$$

However, after the body model was introduced, the matching condition at the target frequency degraded for all separation distances.

Table 1.26: Overall return-loss comparison for the free-space and body-loaded cases.

Case	L (mm)	$f_{\min}$ (GHz)	$S_{11,\min}$ (dB)	$S_{11}(1.120 \text{ GHz})$ (dB)
Free-space tuned	122.0	1.1168	-15.52702	-15.48433
Body 10 mm	122.0	1.0312	-10.49934	-7.839072
Body 5 mm	122.0	0.9792	-11.9744	-5.903831
Body 0.1 mm	122.0	1.3976	-13.8654	-6.951098

For the 10 mm and 5 mm cases, the matching minimum shifted downward in frequency. This is consistent with the expected effect of high-permittivity body loading. The nearby tissue layers increase the effective electrical loading of the dipole, causing the fixed physical length to behave electrically longer than it did in free space.

The 0.1 mm case showed a different behavior. The dominant matching minimum appeared at 1.3976 GHz, which is higher than the target frequency. This result should not be interpreted as a simple resonance shift of an isolated dipole. At this extremely small separation, the dipole and the lossy body model form a strongly coupled electromagnetic system. The input impedance is strongly affected by reactive near-field loading, dielectric loss, finite body geometry, and local field perturbation.

1.8.2 Effect of Body Proximity

The body-separation study demonstrates that antenna performance depends strongly on the distance between the wearable antenna and the body. At 10 mm, the antenna already shows noticeable detuning and mismatch. At 5 mm, the body loading becomes stronger, and the matching minimum shifts further downward to 0.9792 GHz.

The degradation at the target frequency can be summarized as follows:

$$S_{11,\text{free}}(1.120 \text{ GHz}) = -15.48433 \text{ dB},$$

$$S_{11,10\text{mm}}(1.120 \text{ GHz}) = -7.839072 \text{ dB},$$

$$S_{11,5\text{mm}}(1.120 \text{ GHz}) = -5.903831 \text{ dB},$$

and

$$S_{11,0.1\text{mm}}(1.120 \text{ GHz}) = -6.951098 \text{ dB}.$$

These values show that the free-space antenna design cannot be directly used as a wearable antenna without retuning or redesign. The nearby body changes the input impedance sufficiently that the original matching condition is no longer valid.

1.8.3 Efficiency and Gain Degradation

The efficiency and gain results confirm that the lossy body model absorbs part of the accepted electromagnetic power. This is one of the most important effects in wearable antenna design. Even if the antenna has an acceptable S_{11} value at some frequency, a significant portion of the accepted power may be dissipated in the nearby tissue layers instead of being radiated.

Table 1.27: Efficiency and gain comparison at 1.120 GHz.

Case	Radiation Efficiency (dB)	Total Efficiency (dB)	Gain (dBi)
Body 10 mm	-3.410	-4.190	0.7515
Body 5 mm	-3.837	-5.126	0.3534
Body 0.1 mm	-10.52	-11.50	-5.261

The radiation efficiency decreased from -3.410 dB for the 10 mm case to -3.837 dB for the 5 mm case. This reduction is consistent with stronger tissue absorption as the antenna moves closer to the body.

The 0.1 mm case produced a much larger efficiency degradation, with radiation efficiency equal to

$$-10.52 \text{ dB.}$$

This corresponds to severe radiating-performance loss. The gain also dropped to

$$-5.261 \text{ dBi.}$$

This result confirms that extremely small antenna-body spacing is not suitable for efficient radiation unless additional isolation, shielding, matching, or antenna redesign is used.

1.8.4 Near-Field Coupling Versus Far-Field Pattern Evaluation

The body model is located in the near-field region of the dipole for all separation distances used in this project. At 1.120 GHz, the free-space wavelength is

$$\lambda_0 = 267.86 \text{ mm.}$$

Thus,

$$10 \text{ mm} \approx 0.037\lambda_0,$$

$$5 \text{ mm} \approx 0.019\lambda_0,$$

and

$$0.1 \text{ mm} \approx 0.00037\lambda_0.$$

A Python-based wavelength-normalized distance check and selected field-region scale estimates are provided in Appendix [A](#), Listing [A.1](#).

These distances are much smaller than the wavelength, so the body is not a far-field object. It is a nearby loading structure located in the antenna near field.

This does not contradict the use of CST far-field monitors. The CST solver first calculates the electromagnetic fields around the complete antenna-body structure. The far-field monitor then computes the far-zone radiation pattern of that complete structure using a near-to-far-field transformation. Therefore, the body can be in the near field of the antenna, while the reported radiation pattern still represents the far-field radiation behavior of the combined antenna-body system.

1.8.5 Interpretation of the Radiation Patterns

The radiation-pattern results show the combined effect of the dipole antenna and the multilayer body model. In free space, a half-wave dipole is expected to radiate primarily broadside to the wire axis. When the body is introduced, the radiation behavior is modified by two mechanisms.

First, the body changes the current distribution and input impedance of the dipole through near-field coupling. Second, the lossy tissue layers absorb part of the electromagnetic power that would otherwise contribute to radiation. As a result, the gain decreases and the radiation pattern becomes less similar to the ideal free-space dipole pattern.

The 10 mm case retains the most dipole-like radiation behavior among the body-loaded cases because it has the weakest coupling. The 5 mm case shows stronger degradation. The 0.1 mm case shows the most severe effect because the antenna is almost in contact with the tissue model.

1.8.6 Retuning Assessment

The retuning study showed that shortening the dipole can partially compensate for the downward detuning observed in the 5 mm body-separation case. The original 5 mm case had

$$S_{11}(1.120 \text{ GHz}) = -5.903831 \text{ dB}.$$

After shortening the dipole from

$$122.0 \text{ mm}$$

to

106.6 mm,

the target-frequency return loss improved to

$$S_{11}(1.120 \text{ GHz}) = -8.000936 \text{ dB}.$$

This improvement confirms that the original body-loaded antenna was electrically too long. However, the retuned antenna still did not satisfy the common -10 dB matching criterion.

The feed-gap variation also did not provide further improvement. The $L = 106.6$ mm, 1 mm gap case produced

$$S_{11}(1.120 \text{ GHz}) = -7.821004 \text{ dB},$$

which was slightly worse than the 2 mm gap result.

Therefore, the best tested retuned configuration was the $L = 106.6$ mm, 2 mm gap design. This case improved the matching but did not fully restore the free-space performance.

1.8.7 Engineering Implications

The results of this project lead to several important engineering conclusions for wearable antenna design.

First, a dipole antenna tuned in free space cannot be assumed to remain matched when placed near the body. The high-permittivity tissue layers change the effective electrical length and input impedance of the antenna.

Second, impedance matching alone is not sufficient to evaluate wearable antenna performance. Radiation efficiency and gain must also be considered because the body is lossy and can absorb a significant portion of the accepted power.

Third, antenna-body spacing is a critical design parameter. Increasing the separation between the antenna and the body generally reduces near-field coupling and tissue absorption. In practical wearable systems, this can be achieved using a spacer, clothing layer, substrate thickness, ground plane, or antenna structure designed to reduce fields directed into the body.

Fourth, simple length retuning may improve the match but may not be enough for complete performance recovery. A more complete design would likely require additional impedance matching, feed optimization, or a different wearable antenna topology. Examples include planar dipoles, microstrip patch antennas, PIFA-type antennas, backed antennas, or antennas with an artificial magnetic conductor or reflector layer.

1.8.8 Limitations of the Simulation Model

The CST simulations were performed using a simplified multilayer body model and the CST Studio Suite Learning Edition. The model captures the main electromagnetic effect of lossy tissue loading, but it does not represent the full anatomical complexity of the human body.

The finite body patch was used because of the 100,000-cell mesh limitation of the Learning Edition. In addition, the open-boundary spacing was reduced to keep the mesh size within the allowed limit. This introduced a reduced-PML-layer warning in CST. However, the simulations completed successfully and reached the steady-state energy criterion.

Therefore, the results should be interpreted as comparative engineering results rather than as a final medical-grade or regulatory-grade body-interaction study. The conclusions are valid for observing the trends caused by body proximity, dielectric loading, tissue loss, impedance mismatch, and first-order antenna retuning.

1.8.9 Numerical Reliability of the Extreme-Clearance Case

The 0.1 mm separation produced a non-monotonic shift of the dominant matching minimum to 1.3976 GHz. This behavior may reflect strong reactive coupling and a changed impedance branch, but it was not verified through independent mesh refinement or boundary-distance convergence. The Learning Edition cell limit and the intentionally reduced open-boundary spacing prevented a formal convergence campaign. Accordingly, this result is classified as a fixed-model simulation observation. It should not be generalized as a validated monotonic law for near-contact wearable dipoles.

The comparative conclusions that remain directly supported are that the target-frequency match is poor, radiation and total efficiency are strongly reduced, and gain degrades substantially as the antenna approaches the lossy body model. These claims are based on the documented model and are presented without fabrication or measurement validation.

1.8.10 Recommended Future Improvements

A more advanced continuation of this project could include several improvements. The first improvement would be to perform a more systematic optimization of dipole length, feed gap, antenna height, and wire radius. The second improvement would be to add an impedance-matching network to improve S_{11} at 1.120 GHz. The third improvement would be to compare the simple wire dipole with a more realistic wearable antenna geometry, such as a planar dipole or microstrip patch antenna.

A further extension would be to use a higher-resolution body model and evaluate absorbed power or SAR. However, that would require a more detailed anatomical model, a finer mesh, and a simulation setup specifically designed for RF safety assessment. Those tasks are outside the scope of the present project.

1.9 Conclusion

The simulations completed in this project show that the performance of a wearable dipole antenna is strongly dependent on the electromagnetic environment around the antenna. In particular, the simplified multilayer body model significantly changed the impedance matching, efficiency, gain, and radiation behavior compared with the free-space baseline.

The tuned free-space dipole achieved a minimum return loss of

$$S_{11,\min} = -15.52702 \text{ dB}$$

at

$$f_{\min} = 1.1168 \text{ GHz.}$$

At the design operating frequency,

$$f_0 = 1.120 \text{ GHz,}$$

the return loss was

$$S_{11}(1.120 \text{ GHz}) = -15.48433 \text{ dB.}$$

This result confirmed that the tuned free-space dipole provided a valid baseline for studying body-loading effects.

After the multilayer body model was introduced, the antenna performance changed significantly. For the 10 mm separation case, the matching minimum shifted to

$$1.0312 \text{ GHz},$$

and the return loss at 1.120 GHz degraded to

$$-7.839072 \text{ dB}.$$

For the 5 mm separation case, the matching minimum shifted further downward to

$$0.9792 \text{ GHz},$$

and the target-frequency return loss degraded to

$$-5.903831 \text{ dB}.$$

These results show that reducing the antenna-body separation increases the effective dielectric loading of the dipole and causes stronger impedance detuning.

The 0.1 mm case produced the strongest antenna-body interaction. In this case, the dominant matching minimum appeared at

$$1.3976 \text{ GHz},$$

with

$$S_{11,\min} = -13.8654 \text{ dB}.$$

However, the return loss at the target frequency remained poor:

$$S_{11}(1.120 \text{ GHz}) = -6.951098 \text{ dB}.$$

This result was interpreted as a fixed-model response of a strongly coupled lossy antenna-body system rather than a simple monotonic resonance shift of an isolated

half-wave dipole. It was not independently verified through mesh or boundary convergence and therefore is not presented as a convergence-validated physical trend.

The efficiency and gain results also confirmed the strong effect of the body model. At 1.120 GHz, the gain decreased from 0.7515 dBi for the 10 mm case to 0.3534 dBi for the 5 mm case. For the 0.1 mm case, the gain dropped severely to

$$-5.261 \text{ dBi.}$$

The radiation efficiency also decreased significantly, especially for the 0.1 mm case, where CST reported

$$\eta_{\text{rad}} = -10.52 \text{ dB.}$$

This confirms that lossy body tissues absorb a significant portion of the accepted electromagnetic power when the antenna is placed very close to the body.

A retuning study was performed for the 5 mm body-separation case. Since the original 122.0 mm dipole became electrically too long near the body, the dipole was shortened using a first-order frequency-length scaling estimate. The best tested retuned configuration used

$$L = 106.6 \text{ mm}$$

with a

$$2.0 \text{ mm}$$

feed gap. This improved the return loss at 1.120 GHz from

$$-5.903831 \text{ dB}$$

to

$$-8.000936 \text{ dB.}$$

Although this was a clear improvement, the retuned antenna still did not satisfy the common -10 dB matching criterion. This indicates that length-only retuning is not sufficient for complete impedance recovery near the lossy body model.

Overall, the simulation results demonstrate that the human body has a major effect on wearable antenna performance. The body changes the antenna input impedance, shifts the matching frequency, reduces radiation efficiency, lowers gain, and modifies the radiation pattern. The results also show that antenna-body spacing is a critical design parameter. A wearable antenna cannot be designed only in free space and then expected to perform correctly near the body without additional tuning or redesign.

The main engineering conclusion is that practical wearable antenna design requires simultaneous consideration of impedance matching, radiation efficiency, gain, antenna-body separation, and body-induced detuning. For improved performance, future designs may require an impedance-matching network, increased antenna-body spacing, an insulating spacer, feed optimization, or a different antenna topology such as a planar dipole, patch antenna, PIFA-type antenna, or backed wearable antenna structure.

Appendix A

Python Verification Utilities

The script below independently verifies wavelength and length-scaling calculations, electrically normalized body clearances, approximate field-region scales, and conversion of CST efficiency values from decibels to percentages. It does not replace the CST full-wave model.

```
1  """Analytical verification utilities for the wearable half-wave dipole study."""
2
3  from math import sqrt
4
5  C0 = 3.0e8
6  F0_HZ = 1.120e9
7  INITIAL_DIAMETER_MM = 1.81
8  INITIAL_LENGTH_MM = 133.93
9  FREE_SPACE_INITIAL_MIN_GHZ = 1.0200
10 FREE_SPACE_TUNED_LENGTH_MM = 122.0
11 BODY_5MM_MIN_GHZ = 0.9792
12 BODY_RETUNED_LENGTH_MM = 106.6
13 SEPARATIONS_MM = (10.0, 5.0, 0.1)
14
15
16 def db_to_percent(value_db: float) -> float:
17     return 100.0 * 10.0 ** (value_db / 10.0)
18
19
20 def main() -> None:
21     wavelength_mm = C0 / F0_HZ * 1000.0
22     half_wave_mm = wavelength_mm / 2.0
23     radius_mm = INITIAL_DIAMETER_MM / 2.0
24     free_space_estimate_mm = INITIAL_LENGTH_MM * FREE_SPACE_INITIAL_MIN_GHZ /
25     (F0_HZ / 1e9)
26     body_estimate_mm = FREE_SPACE_TUNED_LENGTH_MM * BODY_5MM_MIN_GHZ / (F0_HZ / 1e9)
27     reactive_near_field_mm = 0.62 * sqrt(FREE_SPACE_TUNED_LENGTH_MM**3 /
28     wavelength_mm)
29     far_field_mm = 2.0 * FREE_SPACE_TUNED_LENGTH_MM**2 / wavelength_mm
```

```

28
29 print("Wearable Half-Wave Dipole Verification")
30 print("=====")
31 print(f"Design frequency           : {F0_HZ/1e9:.3f} GHz")
32 print(f"Free-space wavelength       : {wavelength_mm:.2f} mm")
33 print(f"Theoretical half-wave length  : {half_wave_mm:.2f} mm")
34 print(f"Specified wire radius          : {radius_mm:.3f} mm")
35 print(f"Free-space scaling estimate     : {free_space_estimate_mm:.2f} mm")
36 print(f"Selected free-space tuned length : {FREE_SPACE_TUNED_LENGTH_MM:.1f} mm")
37 print(f"5 mm body scaling estimate       : {body_estimate_mm:.2f} mm")
38 print(f"Selected body-retuned length      : {BODY_RETUNED_LENGTH_MM:.1f} mm")
39 print()
40 print("Normalized separations")
41 for value in SEPARATIONS_MM:
42     print(f"    {value:4.1f} mm = {value/wavelength_mm:.6f} lambda0")
43 print(f"Reactive near-field estimate     : {reactive_near_field_mm:.2f} mm")
44 print(f"2D^2/lambda far-field estimate    : {far_field_mm:.2f} mm")
45 print()
46 print("Efficiency conversion from CST dB values")
47 cases = {
48     "10 mm radiation": -3.410,
49     "10 mm total": -4.190,
50     "5 mm radiation": -3.837,
51     "5 mm total": -5.126,
52     "0.1 mm radiation": -10.52,
53     "0.1 mm total": -11.50,
54 }
55 for name, value_db in cases.items():
56     print(f"    {name:18s}: {value_db:7.3f} dB = {db_to_percent(value_db):6.2f}
57 %)
58
59 if __name__ == "__main__":
60     main()

```

Listing A.1: Analytical verification script for the wearable dipole study.

A.1 Console Output

```

Wearable Half-Wave Dipole Verification
=====
Design frequency           : 1.120 GHz
Free-space wavelength       : 267.86 mm
Theoretical half-wave length : 133.93 mm
Specified wire radius      : 0.905 mm
Free-space scaling estimate : 121.97 mm
Selected free-space tuned length : 122.0 mm
5 mm body scaling estimate  : 106.66 mm
Selected body-retuned length : 106.6 mm

```

```
Normalized separations
 10.0 mm = 0.037333 lambda0
  5.0 mm = 0.018667 lambda0
  0.1 mm = 0.000373 lambda0
Reactive near-field estimate      : 51.05 mm
2D^2/lambda far-field estimate   : 111.13 mm

Efficiency conversion from CST dB values
10 mm radiation   : -3.410 dB = 45.60 %
10 mm total      : -4.190 dB = 38.11 %
 5 mm radiation   : -3.837 dB = 41.33 %
 5 mm total      : -5.126 dB = 30.72 %
0.1 mm radiation  : -10.520 dB =  8.87 %
0.1 mm total     : -11.500 dB =  7.08 %
```

Listing A.2: Console output from the wearable dipole verification script.

References

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- [4] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th. Hoboken, NJ, USA: John Wiley & Sons, 2016.